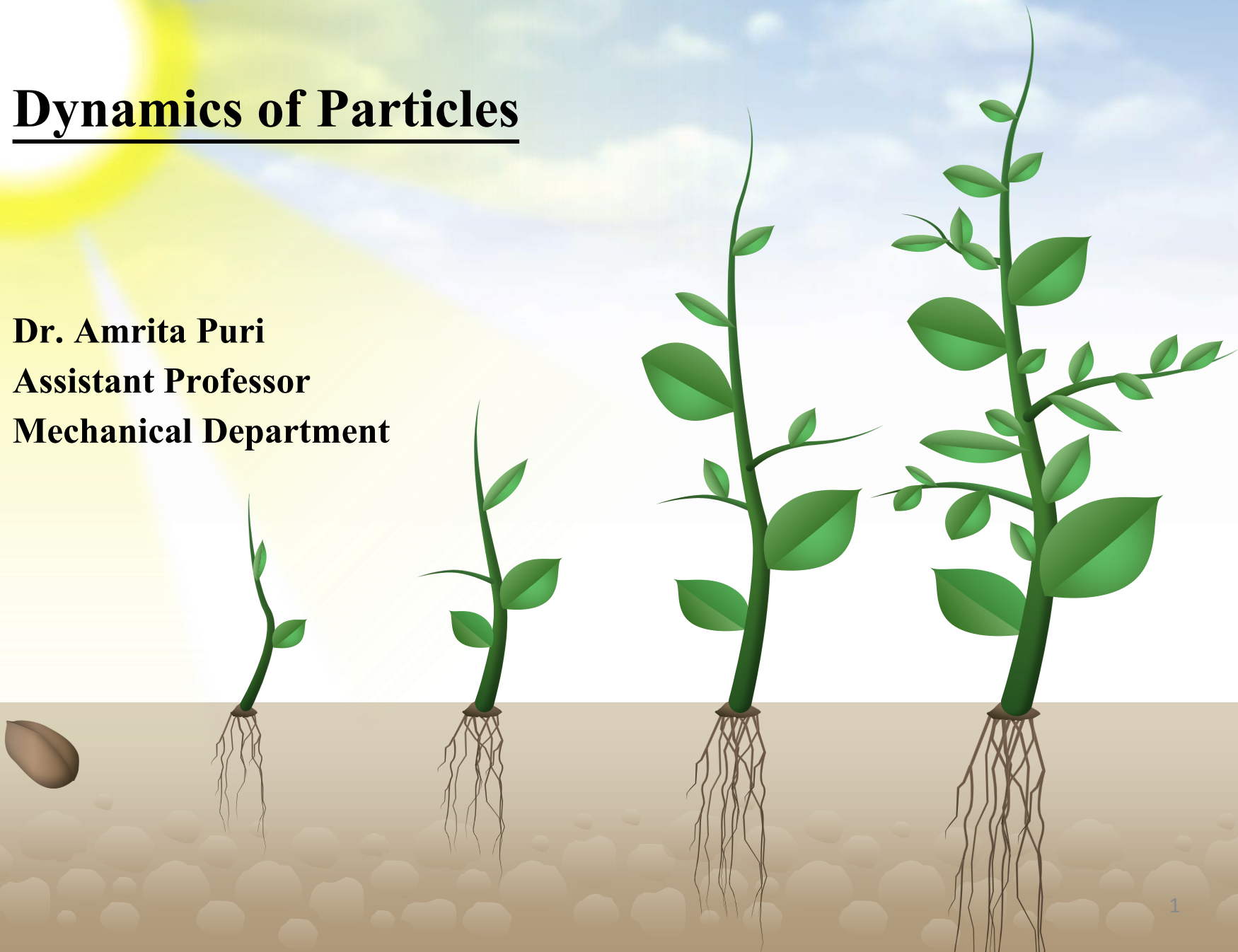


Dynamics of Particles

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Dynamics of Particles

Dynamics of Particles [6 Lectures]:

Chapter 2 [Meriam and Kraige]

Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes

Chapter 3 (single particle) and Chapter 4 (system of particles) [Meriam and Kraige]

Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion, Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles

Kinetics of Particles

- **Kinetics is the study of the relations between unbalanced forces and the resulting changes in motion.**
- Three Approaches to solve Kinetics problems:
 - Direct application of Newton's second law (called the force-mass acceleration method)
 - Work and energy principles
 - Impulse and momentum methods

Primary Inertial System

Primary inertial system (astronomical frame of reference):

Imaginary set of reference axes which are assumed to have no translation or rotation in space.

Newton's Second Law

Experiment performed in primary inertial system

1. A particle of mass, m is subjected to Force F_1 , acceleration a_1 is measured.
2. The particle is subjected to Force F_2 , acceleration a_2 is measured.

Experiment repeated multiple times

$$\frac{F_1}{a_1} = \frac{F_2}{a_2} = \frac{F_3}{a_3} = \dots \dots \dots = C$$

C is a constant and depends upon inertia property of the system.

$$C = km \qquad F = kma$$

Newton's second law

$$F = kma$$

Acceleration is always in the direction of the applied force.

Actual experiment cannot be performed in the ideal manner, the same conclusions have been drawn from countless accurately performed experiments

One of the most accurate checks is given by the precise prediction of the motions of planets based on $F=kma$

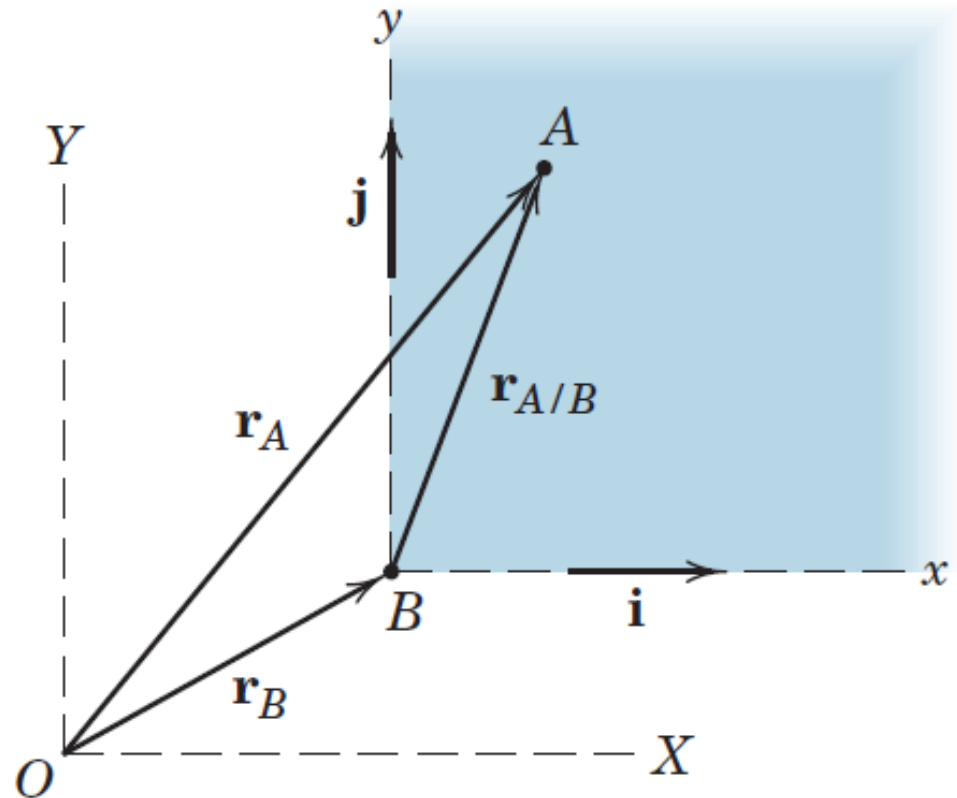
Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

Position vector of A as measured relative to the frame x - y

$$\mathbf{r}_{A/B} = x\mathbf{i} + y\mathbf{j}$$

“ A/B ” means “ A relative to B ”
or “ A with respect to B ”



Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

$$\mathbf{r}_{A/B} = x\mathbf{i} + y\mathbf{j}$$

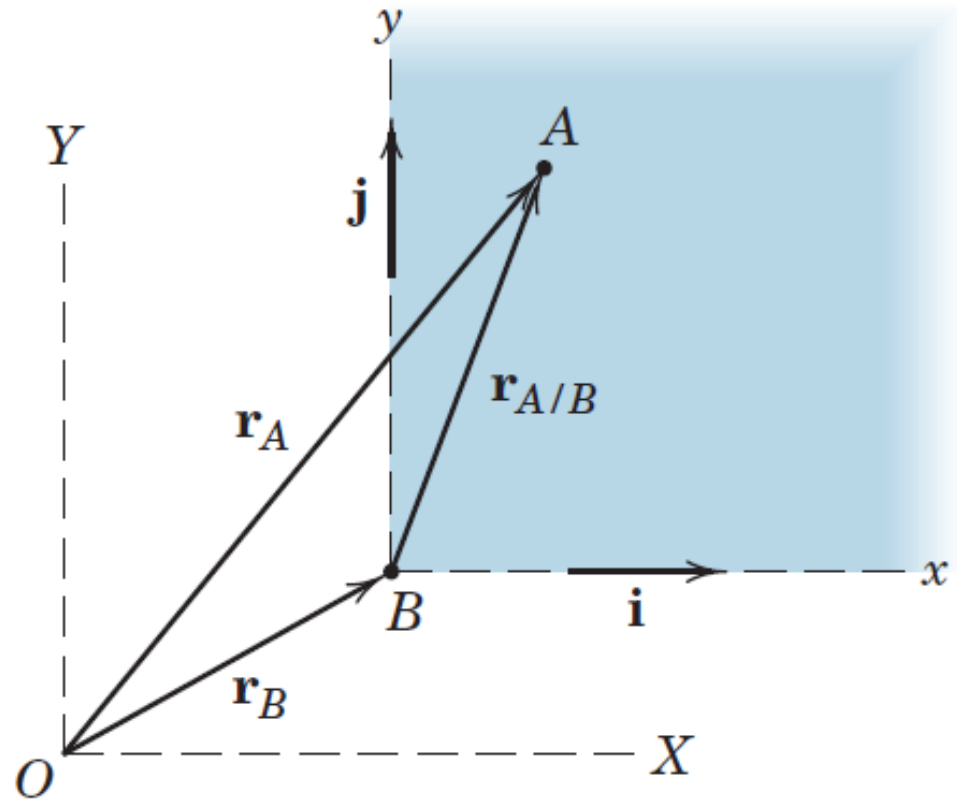
$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}_{A/B}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathbf{r}}_{A/B}$$

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$$

$$\ddot{\mathbf{r}}_A = \ddot{\mathbf{r}}_B + \ddot{\mathbf{r}}_{A/B}$$

$$\mathbf{a}_A = \mathbf{a}_B + \mathbf{a}_{A/B}$$



Inertial System

- Results of the ideal experiment are obtained for measurements made relative to the “fixed” primary inertial system,
Equally valid for measurements made with respect to any nonrotating reference system which translates with a **constant velocity with respect to the primary system.**
- Acceleration measured in a system translating with no acceleration is the same as that measured in the primary system.
- **Newton’s second law holds equally well in a nonaccelerating system, so that we may define an inertial system as any system in which $F=ma$ is valid.**

Task 1

Write an article about ‘Newtonian Mechanics and Theory of relativity’.

System of Units

Kinetic System of Units

k is taken as unity

$$F = ma$$

In kinetic system of units, units for force, mass, and acceleration are not independent.

In SI system of units, units of force (Newton) is derived from units of mass (kg) and units of acceleration (m/sec^2).

Absolute system as units for force depends upon absolute value of mass.

System of Units

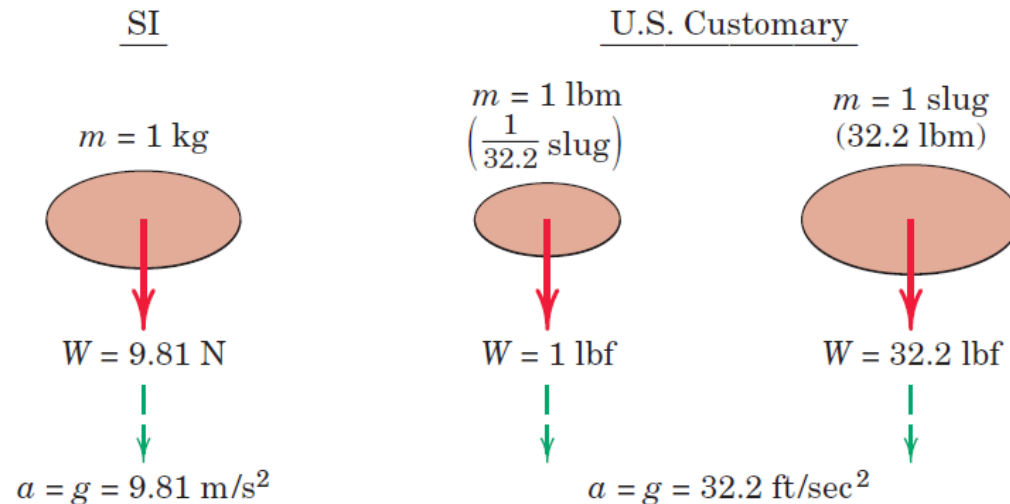
Gravitation system (U.S. Customary unit system)

Units of mass (slugs) is derived from units of force (pound-force, lb) divided by units of acceleration (feet per second squared, ft/sec^2).

Mass is derived from force as determined from gravitational attraction.

Force and Mass Units

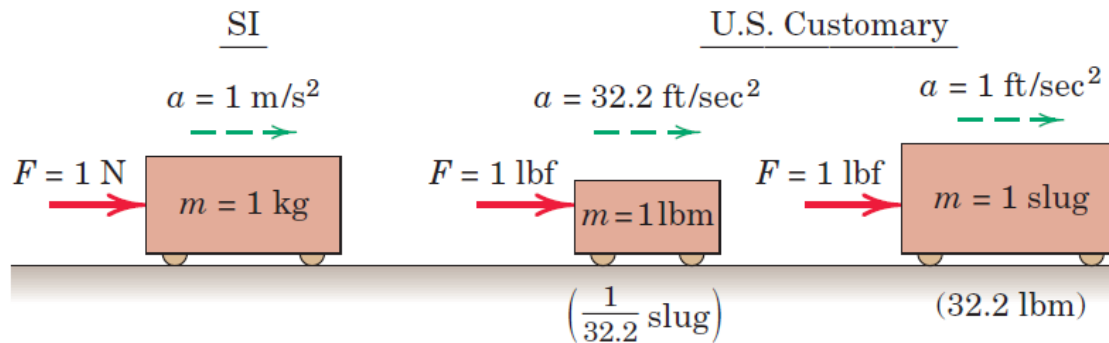
SI system of units	US customary system of units
Mass=1 kg, Weight=9.81 N, $a=g=9.81 \text{ m/sec}^2$	Mass=1 lbm ($1/32.2 \text{ slug}$), Weight=1 lbf, $a=g=32.2 \text{ ft/sec}^2$
	Mass=1 slug (32.2 lbm), Weight=32.2 lbf, $a=g=32.2 \text{ ft/sec}^2$



(a) Gravitational Free-Fall

Force and Mass Units

SI system of units	US customary system of units
Mass=1 kg, Force=1 N, $a=g=1 \text{ m/sec}^2$	Mass=1 lbm (1/32.2 slug), Force=1 lbf, $a=g=32.2 \text{ ft/sec}^2$
	Mass=1 slug (32.2 lbm) Force=1 lbf $a=g=1 \text{ ft/sec}^2$



(b) Newton's Second Law

Equations of motion

A particle of mass m is subjected to the action of concurrent forces $F_1, F_2, F_3, \dots$ whose vector sum is $\sum F$

$$\sum F = ma$$

This equation or any of its component form is called **equation of motion**.

This gives **instantaneous value of acceleration** for **instantaneous value of force** acting on it.

Unconstrained motion and constrained motion

- **Unconstrained motion:** The particle is free of mechanical guides and follows a path determined by its initial motion and by the forces which are applied to it from external sources.

Example: Airplane or rocket in flight and an electron moving in a charged field

- **Constrained motion:** Path of the particle is partially or totally determined by restraining guides.

Example: Train moving along its track and a collar sliding along a fixed shaft

Active Forces: External

Reactive Forces: Reaction forces due to constraining guides.

All the forces, applied and reactive, need to be considered to apply Newton's second law of motion.

Degrees of Freedom

- **Three degrees of freedom:** When the particle moves freely in 3-D space, 3 position coordinates are required to specify its position in space, 3 degrees of freedom.
Example: Airplane or rocket in free flight
- **Two degrees of freedom:** Two coordinates are needed to specify its position in space.
Example: Marble sliding on the surface of a bowl
- **One degree of freedom:** Only one coordinate is required to specify its position in space.
Example: collar moving along a shaft.

Rectilinear motion

- Interested in motion of mass centre of the body
- Body can be assumed as a particle
- Forces can be treated as concurrent through a mass centre
- When motion of the particle can be chosen along one coordinate direction, say, x-direction

$$\sum F_x = m a_x$$

$$\sum F_y = 0$$

$$\sum F_z = 0$$

Rectilinear motion

- Motion of the particle cannot be chosen along one coordinate direction,

$$\sum F_x = m a_x$$

$$\sum F_y = m a_y$$

$$\sum F_z = m a_z$$

$$a^2 = a_x^2 + a_y^2 + a_z^2$$

$$F^2 = F_x^2 + F_y^2 + F_z^2$$

Curvilinear motion

➤ Rectangular Coordinates

$$\sum F_x = ma_x$$

$$a_x = \ddot{x}$$

$$\sum F_y = ma_y$$

$$a_y = \ddot{y}$$

➤ Normal-tangential Coordinates

$$\sum F_n = ma_n$$

$$a_n = v\dot{\beta} = \rho\dot{\beta}^2 = \frac{v^2}{\rho}$$

$$\sum F_t = ma_t$$

$$a_t = \dot{v} = \ddot{s} \quad v = \rho\dot{\beta}$$

➤ Polar Coordinates

$$\sum F_r = ma_r$$

$$a_r = \ddot{r} - r\dot{\theta}^2$$

$$\sum F_\theta = ma_\theta$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

Three Approaches

- **Newton's second law of motion**

$$***F = ma***$$

- **Integration with respect to displacement gives work-energy equation.**
- **Integration with respect to time gives impulse-momentum equation.**

Work

Work done by force during the displacement $d\mathbf{r}$

$$dU = \mathbf{F} \cdot d\mathbf{r}$$

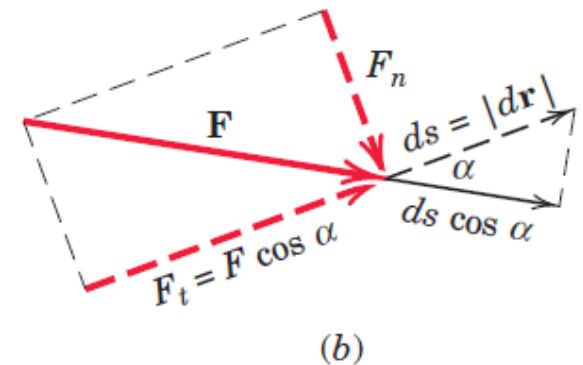
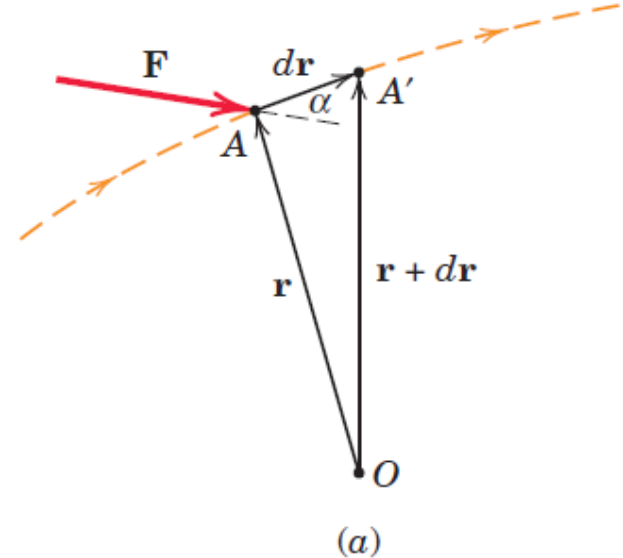
$$dU = F ds \cos \alpha$$

α : angle between $\mathbf{F}$ and $d\mathbf{r}$

ds is magnitude of $d\mathbf{r}$

Interpretations of $dU = F ds \cos \alpha$

- Component of $\mathbf{F}$ in the direction of displacement ($F \cos \alpha$) multiplied by magnitude of displacement
- Component of displacement in the direction of force ($ds \cos \alpha$) multiplied by magnitude of force.



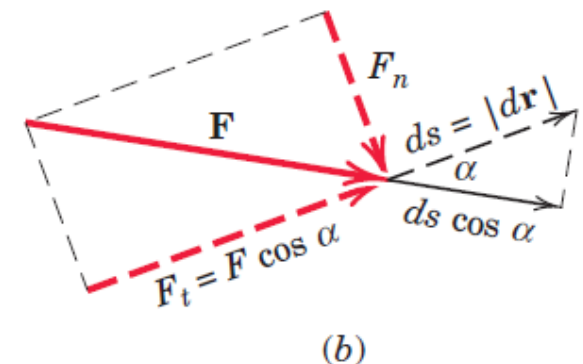
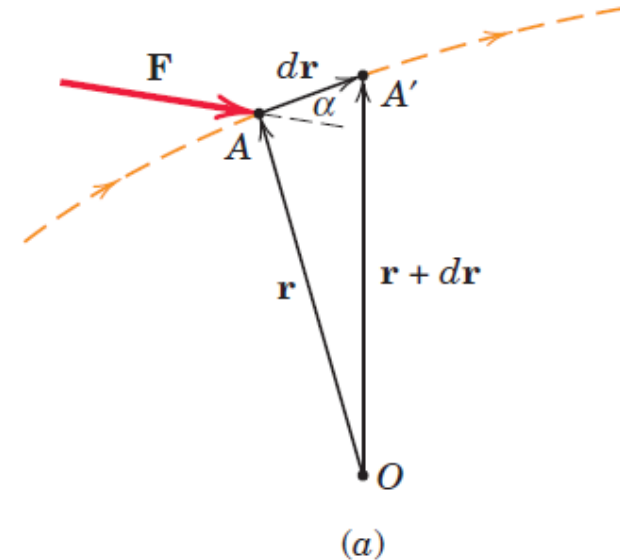
Work

Normal component of the force $\mathbf{F}_n$ does no work.

Work done by force during the displacement $d\mathbf{r}$

$$dU = F_t ds$$

- Work is positive if F_t is in the direction of ds
- Work is negative if F_t is in the opposite of ds
- **Active Forces:** Forces which do work
- **Reactive Forces:** Constraint forces which do no work



Units of Work

1 Joule:

work done by a force of 1 N acting through a distance of 1 m in the direction of the force

Units of Moment or Torque: N-m

Work is scalar quantity.

Moment or Torque is a vector quantity.

In U.S. Customary units

Work (foot-pounds, ft-lb)

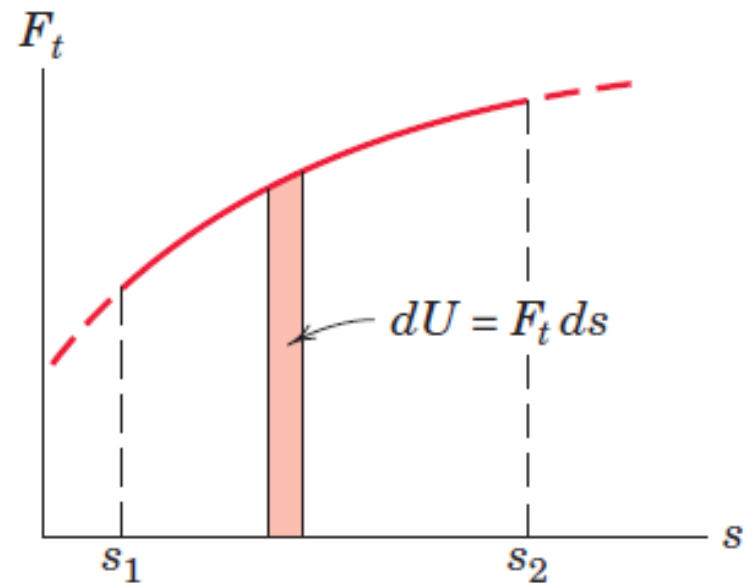
Moment (Pound-feet, lb-ft)

Calculation of Work

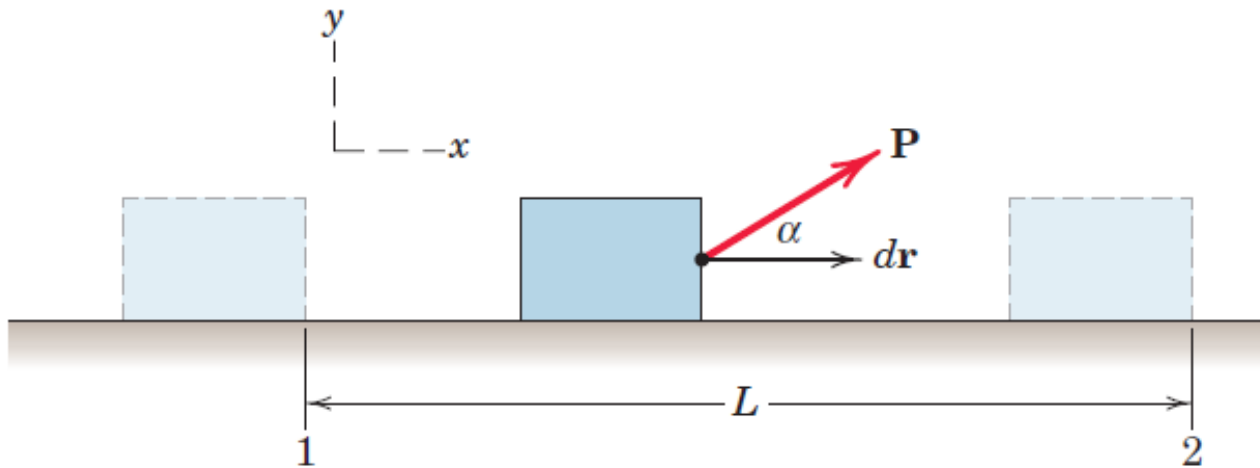
$$U = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_1^2 (F_x dx + F_y dy + F_z dz)$$

$$U = \int_1^2 F_t \cdot ds$$

Work done: Area under $F_t - s$ curve



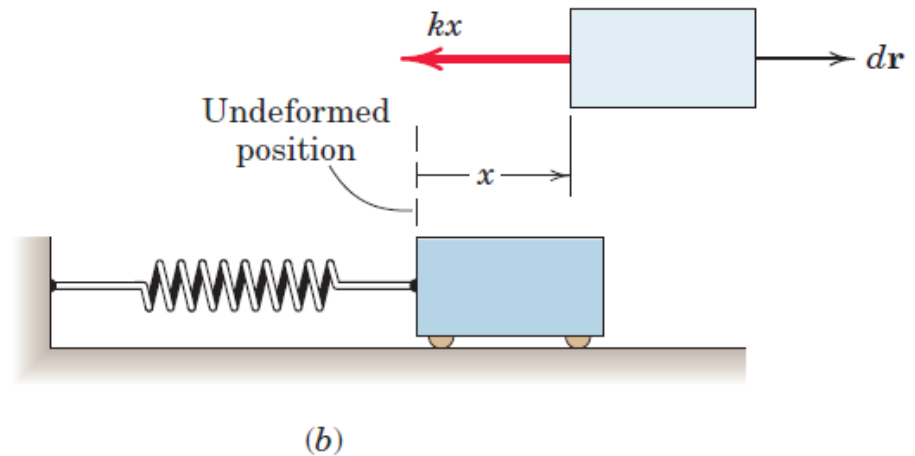
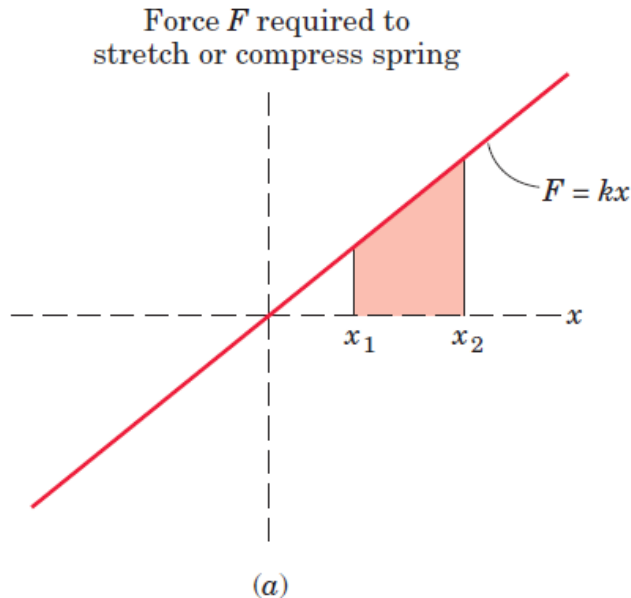
Work Done by a constant force



$$U_{1-2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_1^2 [(P \cos \alpha) \mathbf{i} + (P \sin \alpha) \mathbf{j}] \cdot dx \mathbf{i}$$

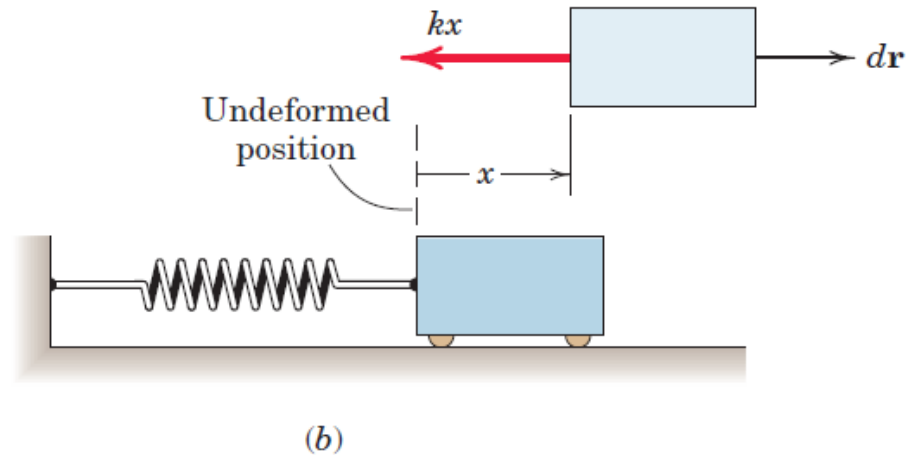
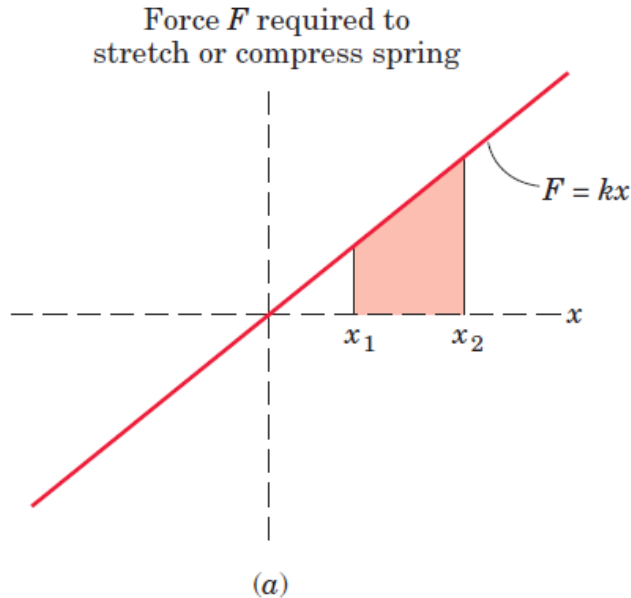
$$U_{1-2} = \int_1^2 (P \cos \alpha) dx = (P \cos \alpha) (x_2 - x_1) = (P \cos \alpha) L$$

Work Done by a spring force



- $F = kx$ is a **linear static relationship**.
- This is valid assumption if **mass of spring is negligible as compared to mass of other parts** in the system.
- Magnitude of work is equal to area under F - x curve.

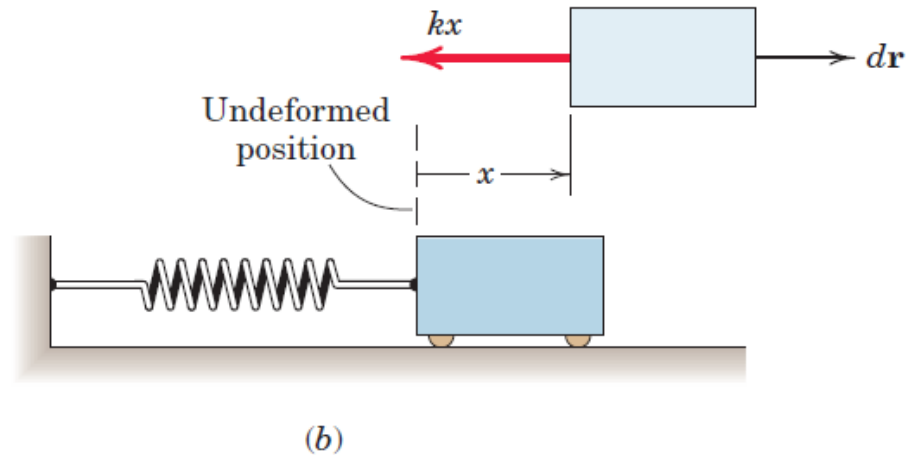
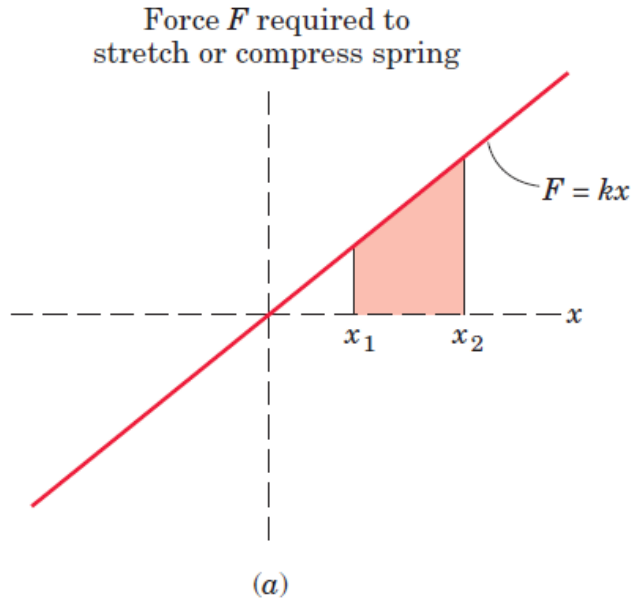
Work Done by a spring force



Linear spring of stiffness k : Force required to stretch or compress the spring is proportional to the deformation x

$$U_{1-2} = \int_1^2 F \cdot dr = \int_1^2 (-kxi) \cdot dxi = \frac{1}{2} k(x_1^2 - x_2^2)$$

Work Done by a spring force



Work done is negative when body moves from undeformed position

Work done is positive when body moves towards undeformed position

$$U_{1-2} = \int_1^2 F \cdot dr = \int_1^2 (-kxi) \cdot dxi = \frac{1}{2} k(x_1^2 - x_2^2)$$

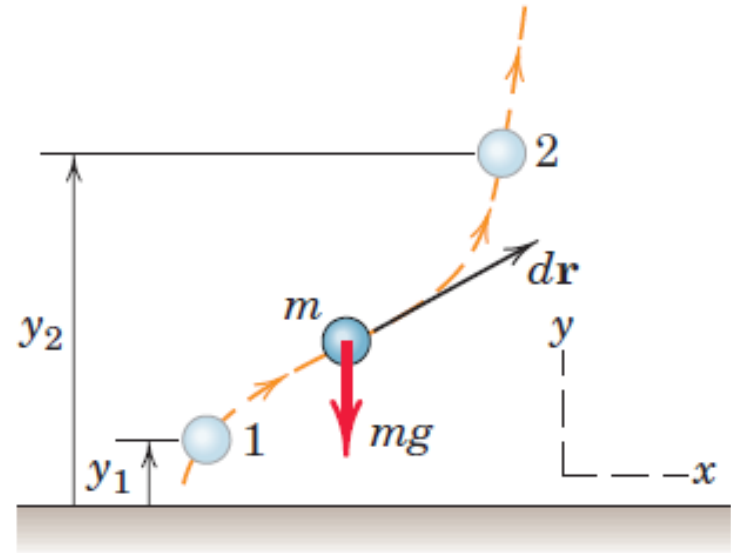
Work Done by Weight

Case (a) Constant Gravity

Altitude variation is sufficiently small.

If body rises, work done is negative.

If body falls, work done is positive.



$$\begin{aligned} U_{1-2} &= \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_1^2 (-mg\mathbf{j}) \cdot (dx\mathbf{i} + dy\mathbf{j}) \\ &= -mg \int_{y_1}^{y_2} dy = -mg(y_2 - y_1) \end{aligned}$$

➤ Horizontal movement does not contribute to this work

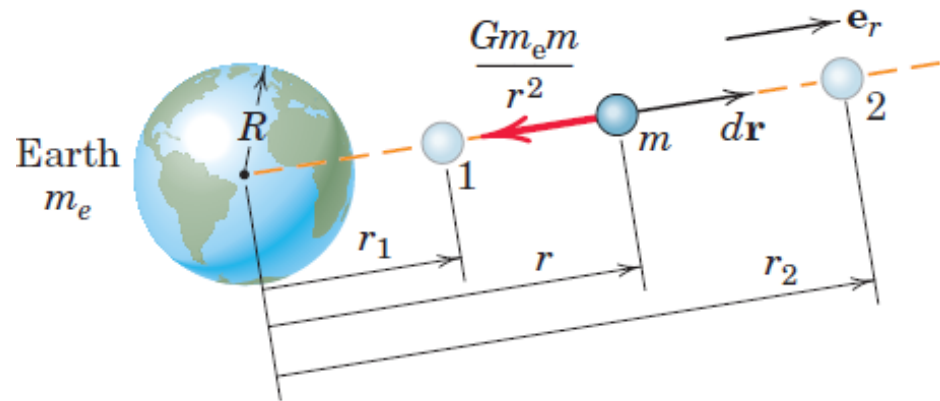
Work Done by Weight

Case (b) Gravity Not Constant

Altitude variation is NOT small.

If body rises, work done is negative.

If body falls, work done is positive.



$$\begin{aligned} U_{1-2} &= \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_1^2 \frac{-Gm_em}{r^2} \mathbf{e}_r \cdot d\mathbf{r} \mathbf{e}_r \\ &= \int_{r_1}^{r_2} \frac{-Gm_em}{r^2} dr = mGm_e \left(\frac{1}{r_2} - \frac{1}{r_1} \right) = mgR^2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right) \end{aligned}$$

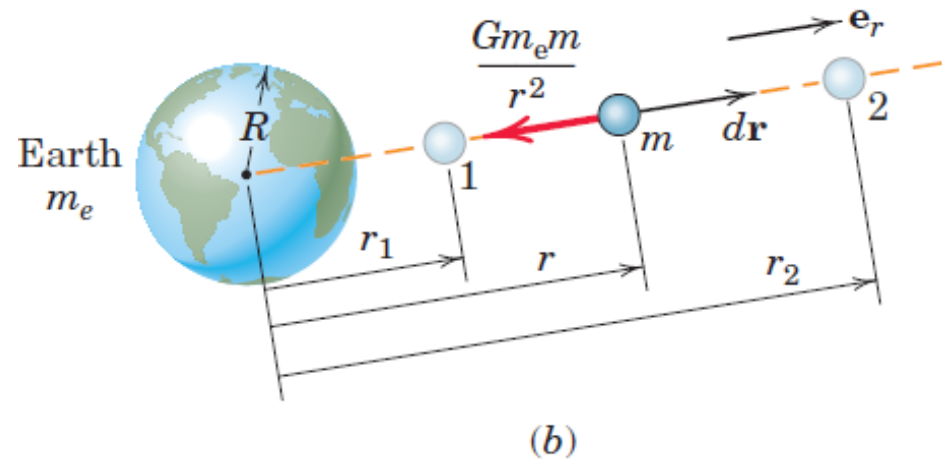
- Transverse movement (perpendicular to radial direction) does not contribute to work

Work Done by Weight

Case (b) Gravity Not Constant

- r represents a radial distance from the center of the earth and not an altitude $h = r - R$ above the surface of the earth

$$U_{1-2} = mgR^2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$$



- Transverse movement (perpendicular to radial direction) does not contribute to work.

Work and Curvilinear motion

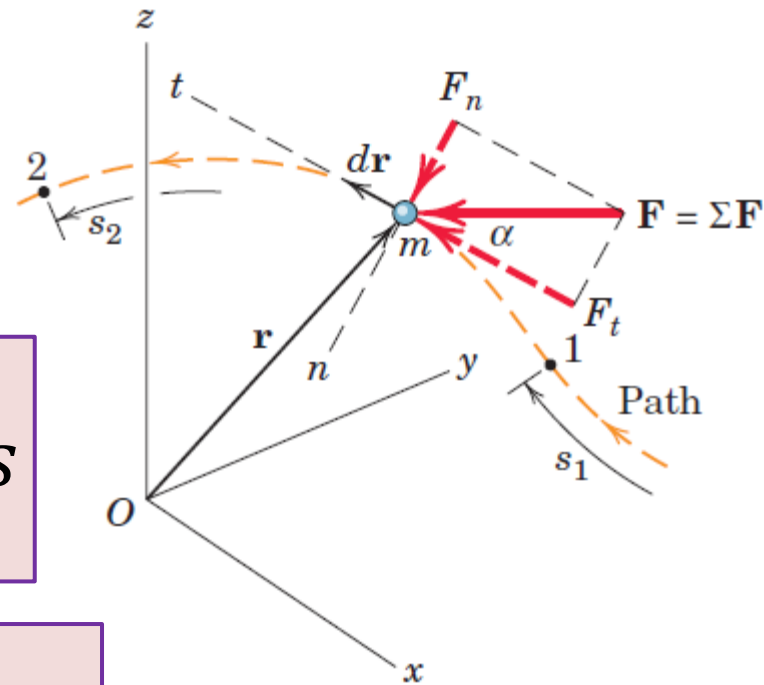
F resultant of all the forces

F acts on the particles of mass m

Particles moves from point 1 to point 2

$$U_{1-2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_{s_1}^{s_2} F_t ds$$

$$U_{1-2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_1^2 m\mathbf{a} \cdot d\mathbf{r}$$

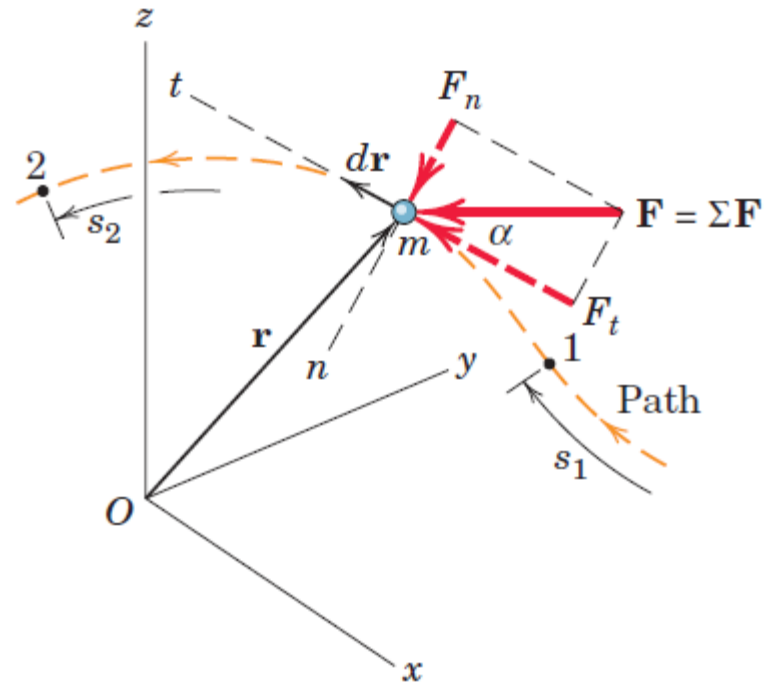


Work and Curvilinear motion

$$U_{1-2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_1^2 m\mathbf{a} \cdot d\mathbf{r}$$

$$\mathbf{a} \cdot d\mathbf{r} = a_t ds$$

$$a_t ds = v dv$$



$$U_{1-2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_{v_1}^{v_2} mv dv = \frac{1}{2} m(v_2^2 - v_1^2)$$

Kinetic Energy

$$U_{1-2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_{v_1}^{v_2} mv \, dv = \frac{1}{2}m(v_2^2 - v_1^2)$$

Kinetic Energy

$$T = \frac{1}{2}m v^2$$

Total work which must be done on the particle to bring it from a state of rest to a velocity v .

Units: Joule (N-m)

Work Energy Equation

$$U_{1-2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_{v_1}^{v_2} mv \, dv = \frac{1}{2}m(v_2^2 - v_1^2)$$

$$U_{1-2} = \frac{1}{2}m(v_2^2 - v_1^2) = T_2 - T_1 = \Delta T$$

Total work done by all forces acting on a particle as it moves from point 1 to point 2 equals the corresponding **change in kinetic energy** of the particle

$$T_1 + U_{1-2} = T_2$$

Advantages of Work Energy Equation

$$U_{1-2} = \frac{1}{2} m(v_2^2 - v_1^2) = T_2 - T_1 = \Delta T$$

$$T_1 + U_{1-2} = T_2$$

- **Avoid the necessity of computing the acceleration** and leads directly to the velocity changes as functions of the forces which do work.
- Only active forces which do work are considered.

Advantages of Work Energy Equation

$$U_{1-2} = \frac{1}{2}m(v_2^2 - v_1^2) = T_2 - T_1 = \Delta T$$

$$T_1 + U_{1-2} = T_2$$

- A system of two particles joined together by a connection which is frictionless and incapable of any deformation
- Internal forces equal and opposite, work done by these forces is equal and cancel out.
- **Only work done by external forces needs to be considered.**

Advantages of Work Energy Equation

$$U_{1-2} = \frac{1}{2} m(v_2^2 - v_1^2) = T_2 - T_1 = \Delta T$$

$$T_1 + U_{1-2} = T_2$$

- U_{1-2} is the total or net work done on the system by forces external to it and ΔT , $T_2 - T_1$, is change in the total kinetic energy of the system.
- The total kinetic energy is the sum of the kinetic energies of both elements of the system.
- It enables us to **analyze a system of particles** joined in the manner described without **dismembering the system**.

Power

Power: Time rate of doing work

Power P developed by a force F which does an amount of work U is,

$$P = \frac{dU}{dt} = \frac{F \cdot d\mathbf{r}}{dt} = F \cdot \mathbf{v}$$

$\frac{d\mathbf{r}}{dt}$ is the velocity $\mathbf{v}$ of the point of application of the force.

Units of Power

Units of Power: $\text{N}\cdot\text{m}/\text{sec} = \text{Joule}/\text{sec} = \text{Watt}$

1 Watt equals one joule per second (J/s).

In U.S. customary units,

Unit for mechanical power is *horsepower* (hp).

$$\mathbf{1\ Watt = 1\ J/s}$$

$$\mathbf{1\ horsepower = 550\ ft\cdot lb/sec = 33000\ ft\cdot lb/min}$$

$$\mathbf{1\ horsepower = 746\ W = 0.746\ kW}$$

Efficiency

Mechanical efficiency e_m of the machine:

Ratio of the work done *by* a machine to the work done *on* the machine during the same time interval

$$e_m = \frac{P_{output}}{P_{input}}$$

$$e = e_m e_t e_e$$

e : Overall efficiency

e_t : thermal efficiency

e_e : electrical efficiency

Gravitational Potential Energy

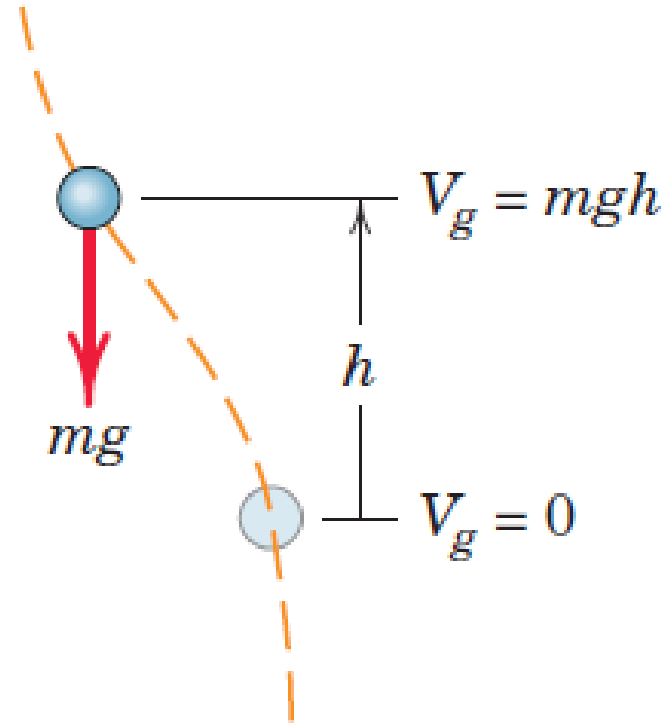
Gravitational potential energy, V_g , of the particle:

Work mgh done against the gravitational field to elevate the particle a distance h above some arbitrary reference plane (called a datum), where V_g is taken to be zero

$$V_g = mgh$$

In going from one level at $h = h_1$ to a higher level at $h = h_2$, the *change* in potential energy becomes

$$\Delta V_g = mg(h_2 - h_1) = mg\Delta h$$



Gravitational Potential Energy

- In going from one level at $h = h_1$ to a higher level at $h = h_2$, the *change* in potential energy becomes

$$\Delta V_g = mg(h_2 - h_1) = mg\Delta h$$

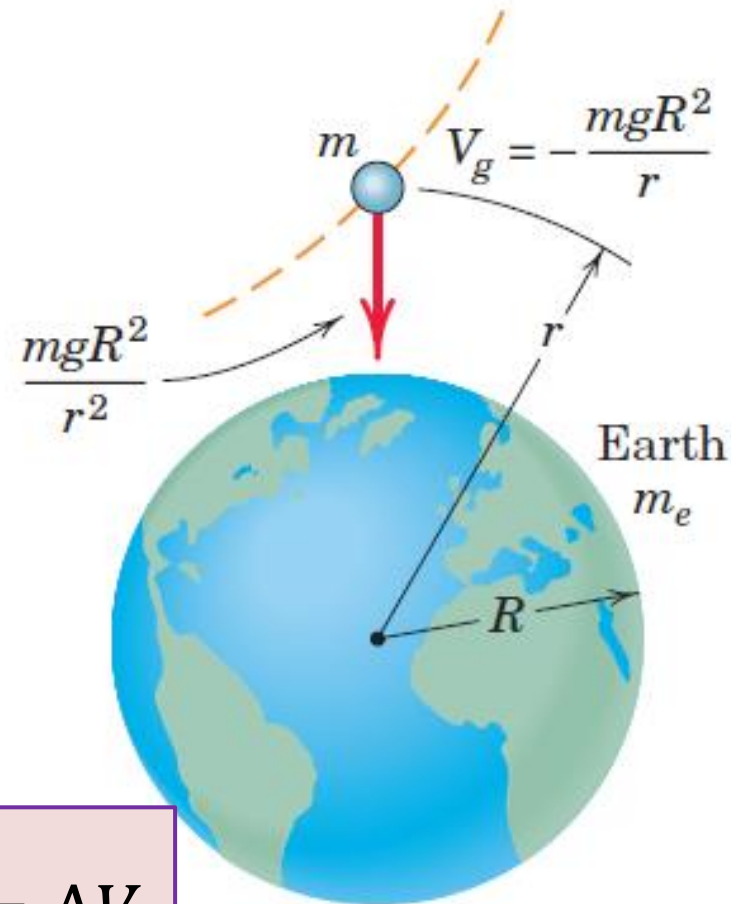
- The corresponding **work done by the gravitational force on the particle** is $-mg\Delta h$.
- The **work done by the gravitational force** is the **negative of the change in potential energy**.

Gravitational Potential Energy

$$F = \frac{Gmm_e}{r^2} = \frac{mgR^2}{r^2}$$

Work done *against* this force to change the radial position of the particle from r_1 to r_2 is the change $[(V_g)_2 - (V_g)_1]$ in gravitational potential energy

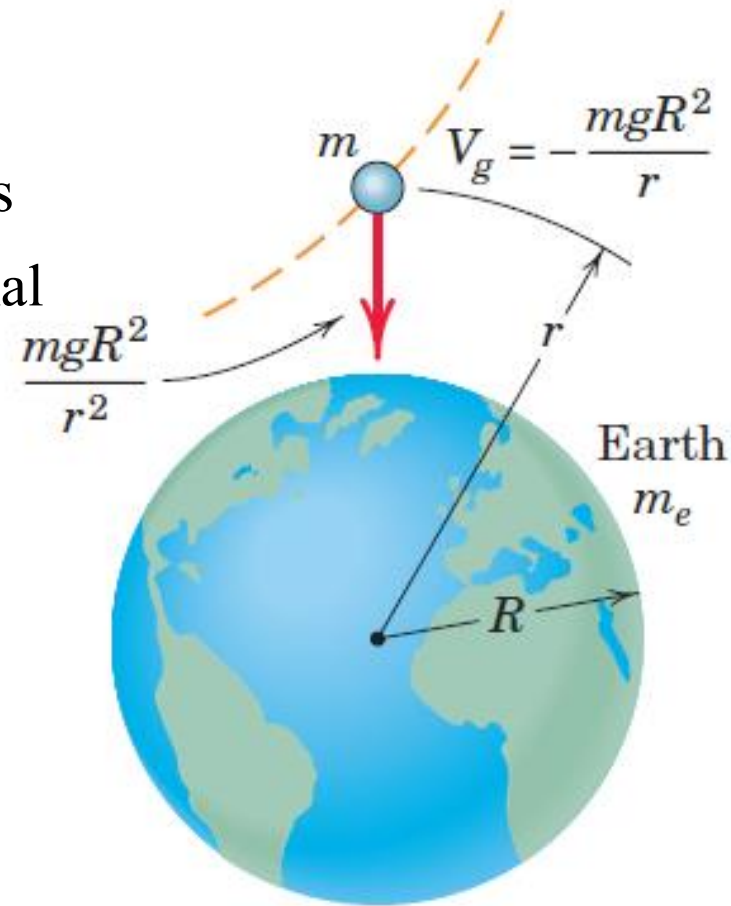
$$\int_{r_1}^{r_2} mgR^2 \frac{dr}{r^2} = mgR^2 \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \Delta V_g$$



Gravitational Potential Energy

Work done *against* this force to change the radial position of the particle from r_1 to r_2 is the change $[(V_g)_2 - (V_g)_1]$ in gravitational potential energy.

$$\Delta V_g = mgR^2 \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$



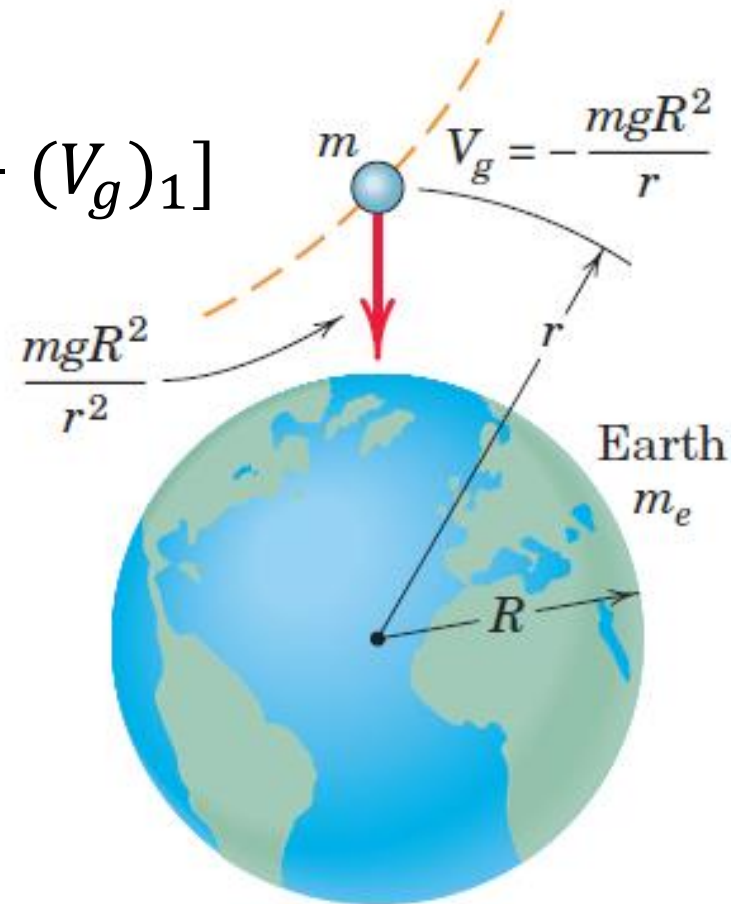
ΔV_g is the *negative* of the work done *by* the gravitational force.

Gravitational Potential Energy

$$\Delta V_g = mgR^2 \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = [(V_g)_2 - (V_g)_1]$$

$(V_g)_2 = 0$ when r_2 tends to infinity,

$$V_g = - \left(\frac{mgR^2}{r} \right)$$



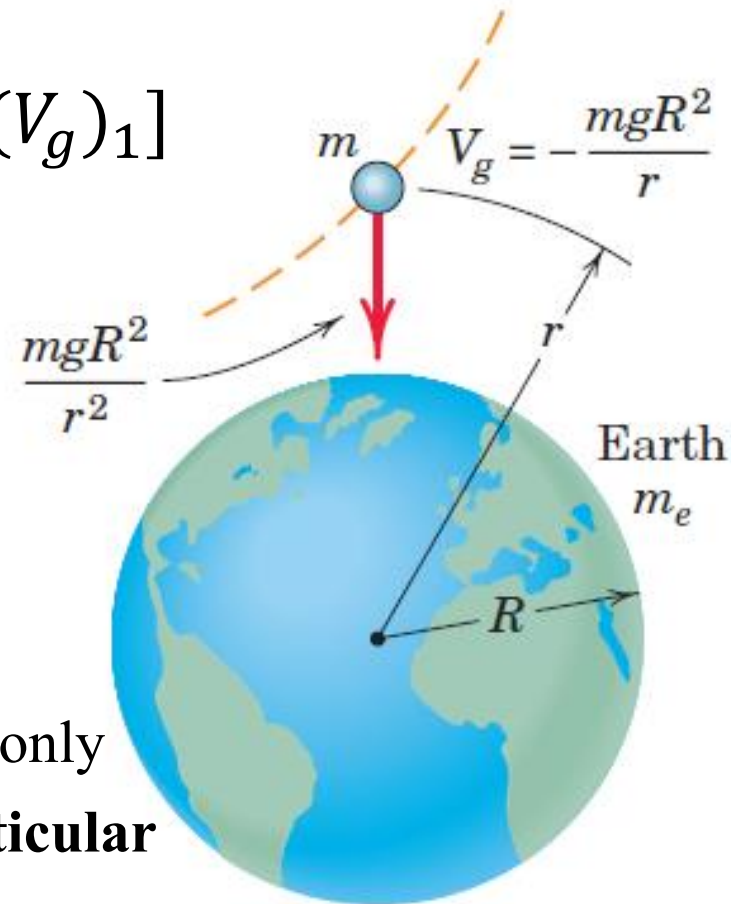
Gravitational Potential Energy

$$\Delta V_g = mgR^2 \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = [(V_g)_2 - (V_g)_1]$$

$(V_g)_2 = 0$ when r_2 tends to infinity,

$$V_g = - \left(\frac{mgR^2}{r} \right)$$

Potential energy of a given particle depends only on **its position**, h or r , and **NOT on the particular path** it followed in reaching that position.



Elastic Potential Energy

- **Work done, on the spring to deform it**, is stored in the spring and is called its **elastic potential energy** V_e
- During the release of the deformation of the spring, energy is recoverable in the form of work done by the spring on the body attached to its movable end

$$V_e = \int_0^x kx \, dx = \frac{1}{2} kx^2$$

$$\Delta V_e = \frac{1}{2} k(x_2^2 - x_1^2)$$

Elastic Potential Energy

$$\Delta V_e = \frac{1}{2} k (x_2^2 - x_1^2)$$

- **Deformation**, either tensile or compressive, of a spring **increases** from x_1 to x_2 during the motion, **change in potential energy of the spring is positive**
- **Deformation**, either tensile or compressive, of a spring **decreases** from x_1 to x_2 during the motion, **change in potential energy of the spring is negative**

Elastic Potential Energy

- Force exerted on the spring by the moving body is equal and opposite to the force F exerted by the spring on the body
- Work done on the spring is the negative of the work done on the body.
- Work U done by the spring on the body, the negative of the potential energy change for the spring $-\Delta V_e$
$$U = -\Delta V_e$$

Work-Energy Equation

$$U_{1-2} = T_2 - T_1 = \Delta T$$

$$U'_{1-2} + (-\Delta V_g) + (-\Delta V_e) = \Delta T$$

$$U'_{1-2} = \Delta T + \Delta V$$

$$\Delta V = (\Delta V_g) + (\Delta V_e)$$

ΔV_g depends upon end point positions of the particle

ΔV_e depends upon end point lengths of elastic spring

$$T_1 + V_1 + U'_{1-2} = T_2 + V_2$$

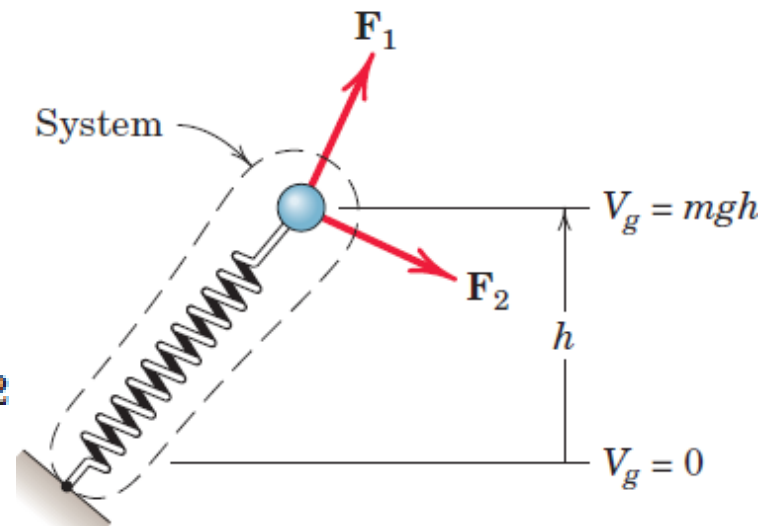
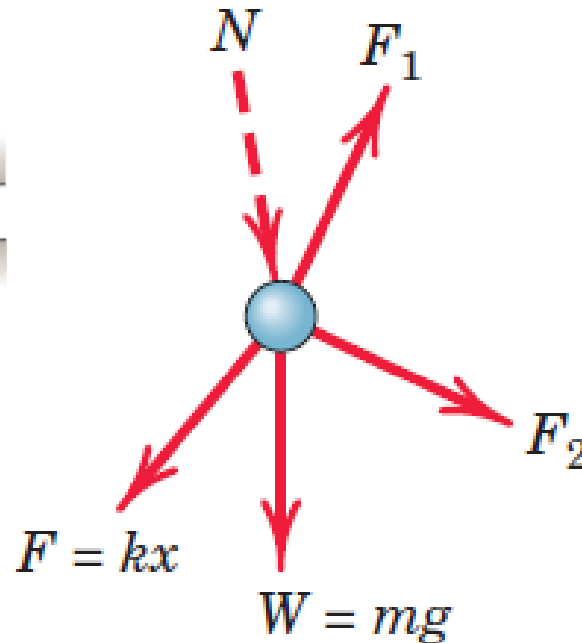
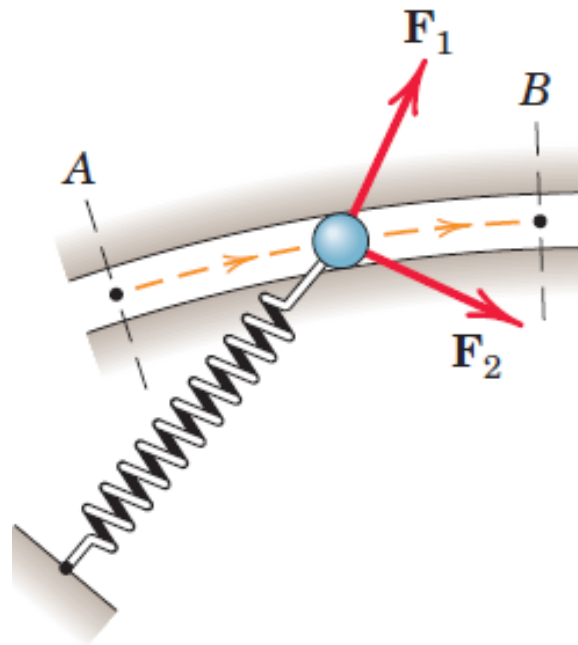
Work-Energy Equation

$$U_{1-2} = T_2 - T_1 = \Delta T$$

$$U'_{1-2} + (-\Delta V_g) + (-\Delta V_e) = \Delta T$$

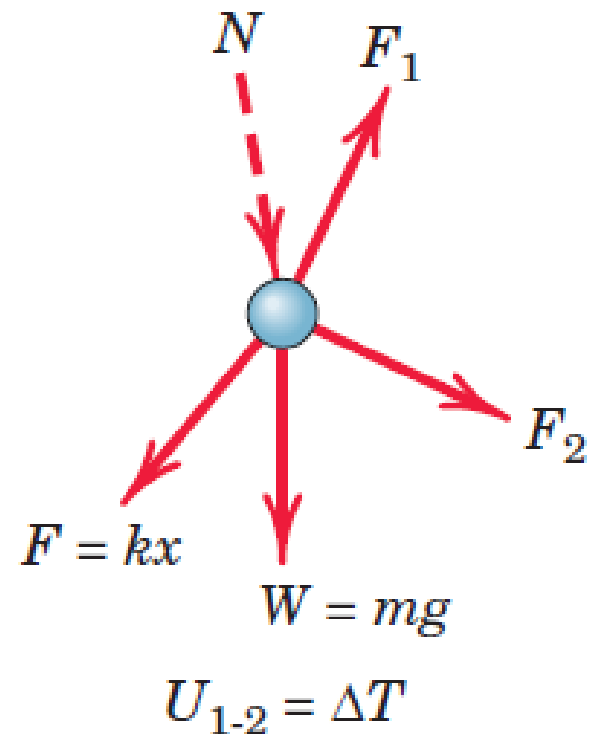
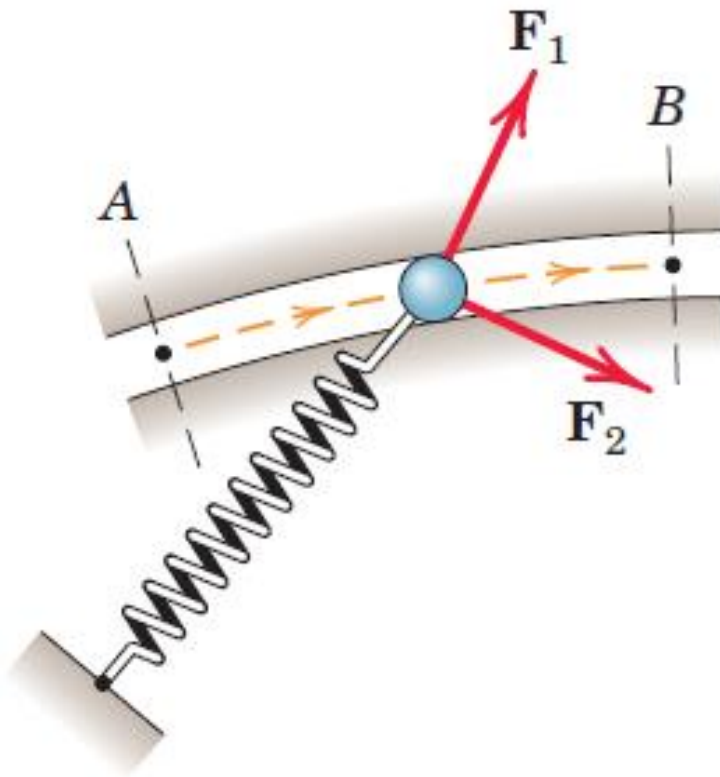
$$U_{1-2} = \Delta T$$

$$U'_{1-2} = \Delta T + \Delta V$$



Work-Energy Equation

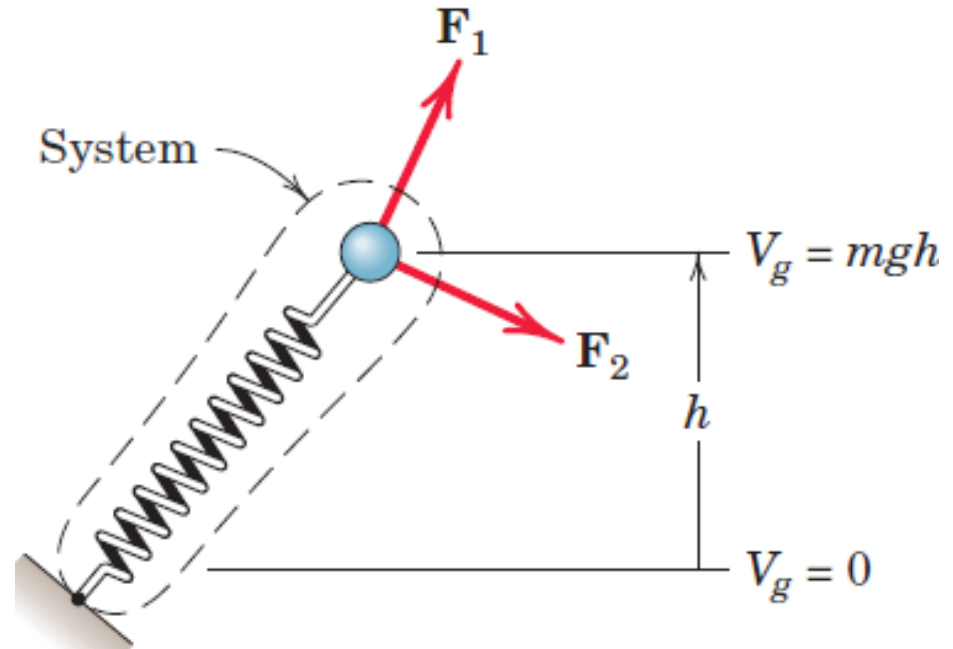
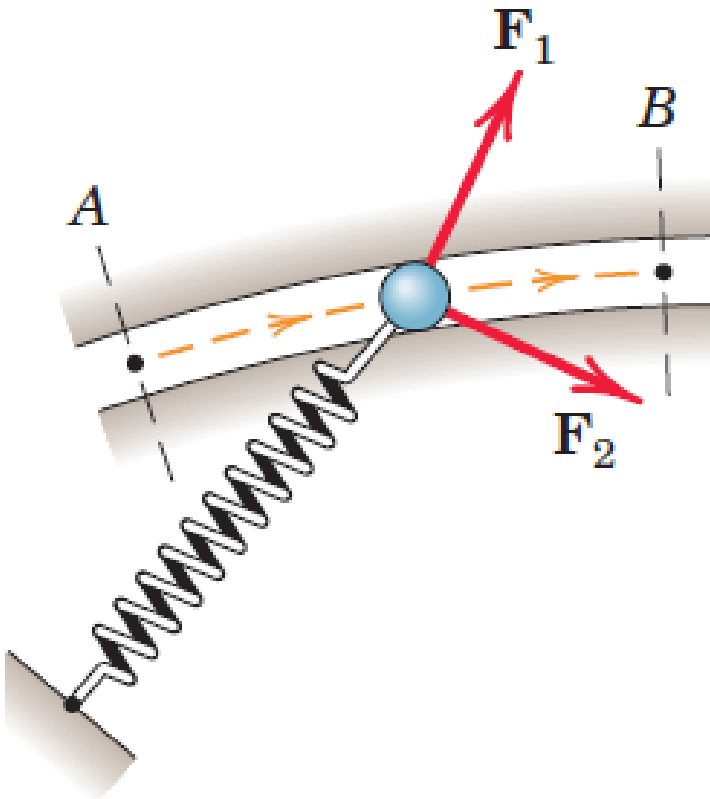
$$U_{1-2} = T_2 - T_1 = \Delta T$$



Work-Energy Equation

$$U'_{1-2} + (-\Delta V_g) + (-\Delta V_e) = \Delta T$$

$$U'_{1-2} = \Delta T + \Delta V$$



Law of conservation of dynamical energy

$$U'_{1-2} + (-\Delta V_g) + (-\Delta V_e) = \Delta T$$

$$U'_{1-2} = \Delta T + \Delta V$$

NO external force act on the particle

If particle is subjected to only spring force, gravitational force and constraint forces, $U'_{1-2} = 0$

$$\Delta T + \Delta V = 0$$

$$T_2 - T_1 + V_2 - V_1 = 0$$

$$T_1 + V_1 = T_2 + V_2$$

$$E_1 = E_2$$

Law of conservation of dynamical energy

$$U'_{1-2} + (-\Delta V_g) + (-\Delta V_e) = \Delta T$$

$$U'_{1-2} = \Delta T + \Delta V$$

$$T_1 + V_1 = T_2 + V_2$$

$$E_1 = E_2$$

When E is constant, transfers of energy between kinetic and potential may take place as long as the total mechanical energy $T + V$ does not change

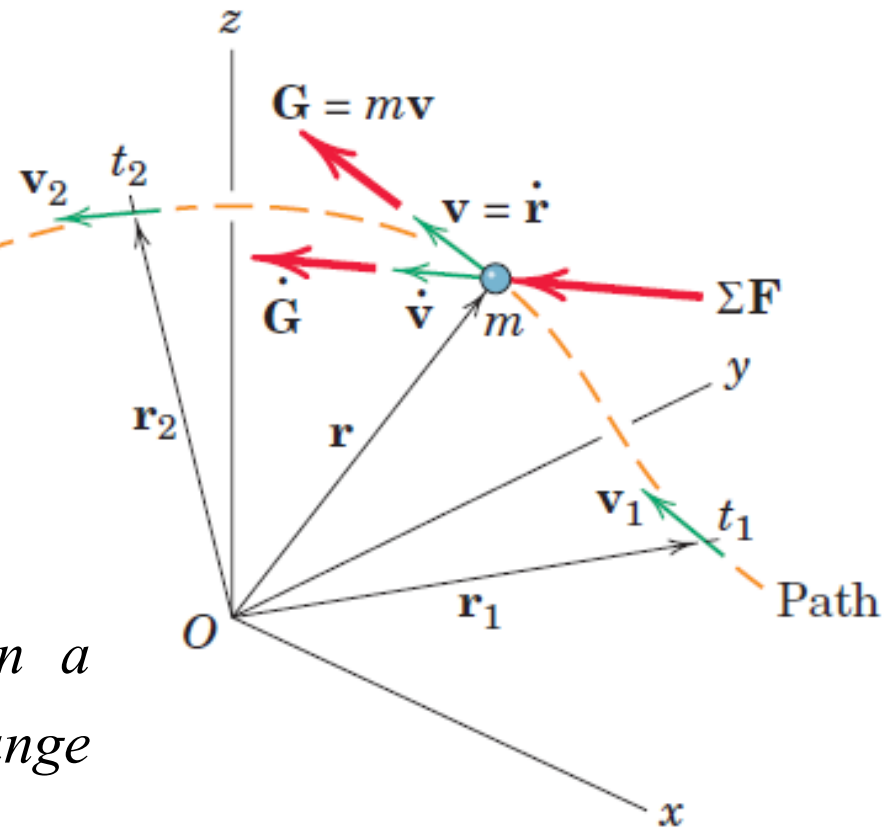
Linear Momentum

Linear momentum of the particle, $\mathbf{G}$: Product of the mass and velocity

$$\sum \mathbf{F} = m\dot{\mathbf{v}} = \frac{d}{dt}(m\mathbf{v})$$

$$\sum \mathbf{F} = \dot{\mathbf{G}}$$

Resultant of all forces acting on a particle equals its time rate of change of linear momentum.



Units of Linear Momentum

SI System: kg. m-sec, N-sec

US Customary units: [lb/(ft/sec²)] [ft/sec]: lb-sec

$$\sum F_x = \dot{G}_x$$

$$\sum F_y = \dot{G}_y$$

$$\sum F_z = \dot{G}_z$$

Linear Impulse and Momentum Principle

$$\sum F = m\dot{v} = \frac{d}{dt}(mv)$$

$$\sum F = \dot{G}$$

$$\int_{t_1}^{t_2} \sum F dt = G_2 - G_1 = \Delta G$$

- **Linear impulse of the force:** Product of force and time
- **Total linear impulse on m equals the corresponding change in linear momentum of m**

Impulse-Momentum Principle

Initial linear momentum of the body plus the linear impulse applied to it equals its final linear momentum

$$G_1 + \int_{t_1}^{t_2} \sum F \, dt = G_2$$

$$mv_{1x} + \int_{t_1}^{t_2} \sum F_x \, dt = mv_{2x}$$

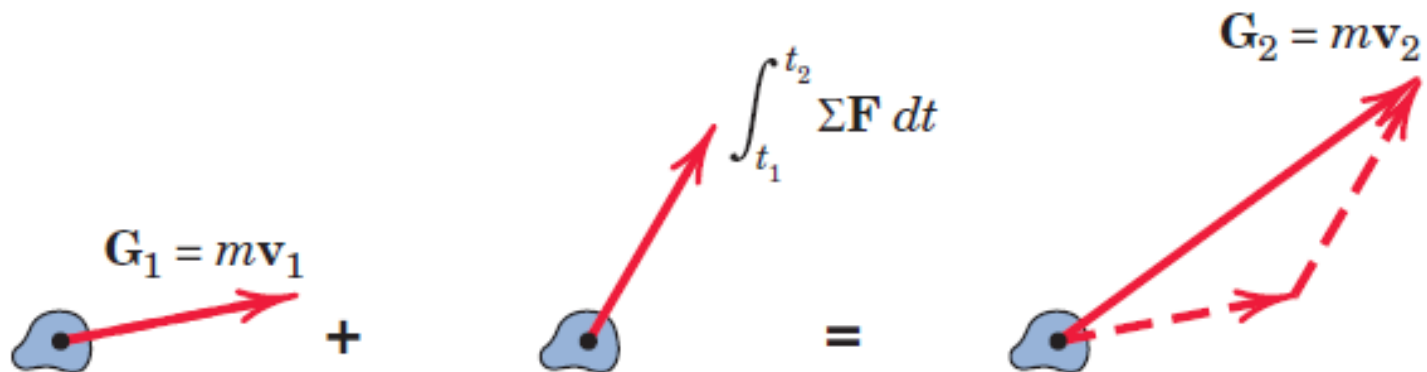
$$mv_{1y} + \int_{t_1}^{t_2} \sum F_y \, dt = mv_{2y}$$

$$mv_{1z} + \int_{t_1}^{t_2} \sum F_z \, dt = mv_{2z}$$

Impulse-Momentum Diagram

$$G_1 + \int_{t_1}^{t_2} \sum F dt = G_2$$

Initial linear momentum of the body plus the linear impulse applied to it equals its final linear momentum



Impulsive and Non-impulsive Forces

Impulsive Forces: Forces are very large and of short duration

Example: Forces of short impact

Non-impulsive Forces

Example: Weight of ball

In many cases, non-impulsive forces may be negligible as compared to impulsive forces

Weight of the ball (about 5 oz) is small compared with the force (which could be several hundred pounds in magnitude) exerted on the ball by the bat

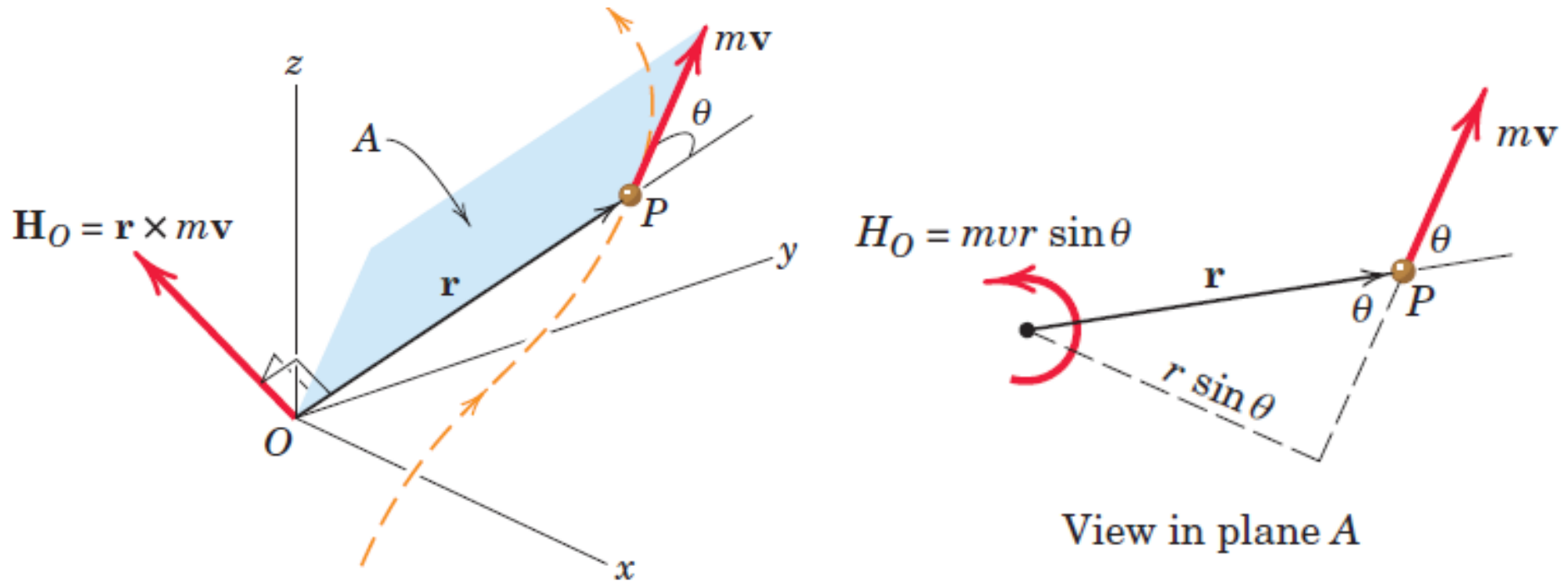


Conservation of Linear Momentum

$$\int_{t_1}^{t_2} \sum F \, dt = G_2 - G_1 = \Delta G = 0$$
$$G_1 = G_2$$

- If the **resultant force** on a particle is **zero** during an interval of time, its **linear momentum** G remains **constant**
- Linear momentum of the particle is said to be conserved
- Total linear impulse on the particle in a **particular direction** is zero, the **corresponding linear momentum** is unchanged (conserved) in that **direction**.

Angular Momentum



Angular Momentum: moment of linear momentum

The angular momentum then is a vector perpendicular to the plane A defined by $\mathbf{r}$ and $\mathbf{v}$.

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v}$$

$$H_O = mvr \sin \theta$$

Angular Momentum

Angular Momentum: moment of linear momentum

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v}$$

$$\mathbf{H}_O = m \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ v_x & v_y & v_z \end{bmatrix}$$

$$\mathbf{H}_O = m(yv_z - zv_y)\mathbf{i} + m(zv_x - xv_z)\mathbf{j} + m(xv_y - yv_x)\mathbf{k}$$

Angular Momentum

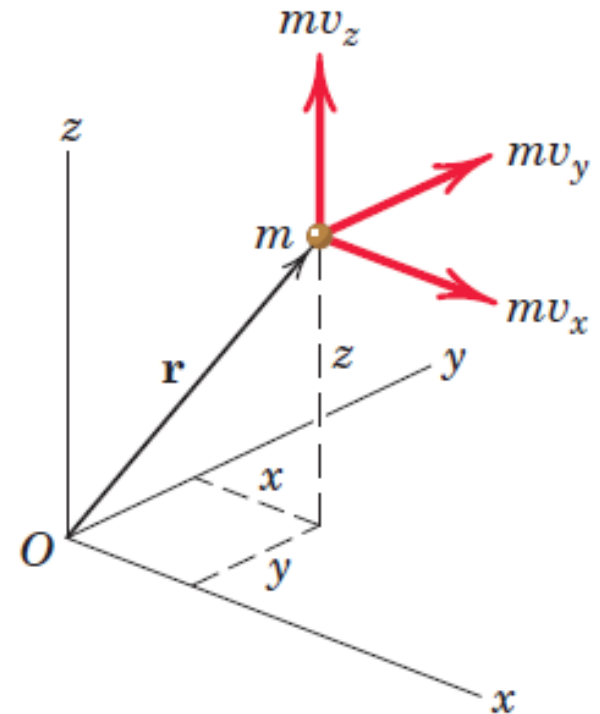
Angular Momentum: moment of linear momentum

$$\mathbf{H}_O = m (y v_z - z v_y) \mathbf{i} + m (z v_x - x v_z) \mathbf{j} + m (x v_y - y v_x) \mathbf{k}$$

$$H_x = m (y v_z - z v_y)$$

$$H_y = m (z v_x - x v_z)$$

$$H_z = m (x v_y - y v_x)$$



Units of Angular Momentum

SI Units of angular momentum

(Kg.m/s).m= N.m.s

$$H_o = r \times mv$$

US Customary Units

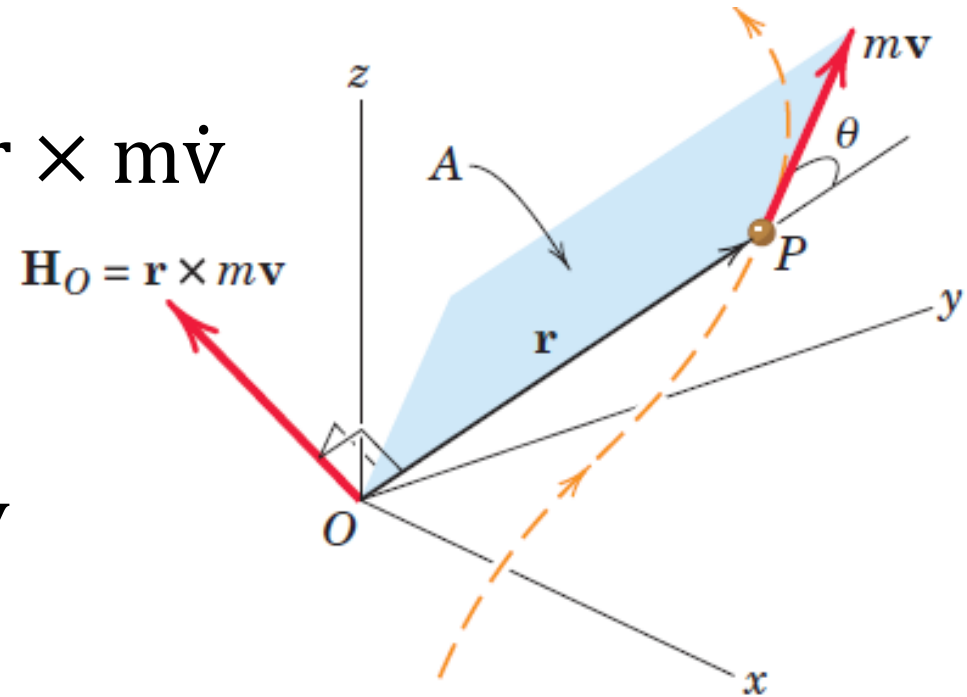
[lb/(ft/sec²)][ft/sec][ft]=lb-ft-sec

Rate of change of Angular Momentum

$$\sum M_o = \mathbf{r} \times \sum \mathbf{F} = \mathbf{r} \times m\dot{\mathbf{v}}$$

$$\dot{\mathbf{H}}_o = \dot{\mathbf{r}} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}}$$

$$\dot{\mathbf{H}}_o = \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}} = 0 + \mathbf{r} \times m\dot{\mathbf{v}} = \sum M_o$$



Rate of change of Angular Momentum

$$\sum M_o = \mathbf{r} \times \sum \mathbf{F} = \mathbf{r} \times m\dot{\mathbf{v}}$$

$$\dot{H}_o = \dot{\mathbf{r}} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}}$$

$$\dot{H}_o = \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}} = 0 + \mathbf{r} \times m\dot{\mathbf{v}} = \sum M_o$$

$$\dot{H}_o = \sum M_o$$

Moment about the fixed point O of all forces acting on m equals the time rate of change of angular momentum of m about O.

Rate of change of Angular Momentum

$$\sum \mathbf{M}_O = \dot{\mathbf{H}}_O$$

$$\sum M_{Ox} = \dot{H}_{Ox}$$

$$\sum M_{Oy} = \dot{H}_{Oy}$$

$$\sum M_{Oz} = \dot{H}_{Oz}$$

Moment about the fixed point O of all forces acting on m equals the time rate of change of angular momentum of m about O.

Angular Impulse-Momentum Principle

Angular impulse: Product of moment and time

$$\sum M_o = \dot{H}_o \qquad \sum M_o dt = dH_o$$

$$\int_{t_1}^{t_2} \sum M_o dt = \Delta H_o = (H_o)_2 - (H_o)_1$$

Total angular impulse on m about the fixed point O equals the **corresponding change in angular momentum of m about O**.

$$(H_o)_2 = \mathbf{r} \times m\mathbf{v}_2 \qquad (H_o)_1 = \mathbf{r} \times m\mathbf{v}_1$$

Angular Impulse-Momentum Principle

Initial angular momentum of the particle plus the angular impulse applied to it equals its final angular momentum

$$(H_o)_1 + \int_{t_1}^{t_2} \sum M_o dt = (H_o)_2$$

$$(H_{ox})_1 + \int_{t_1}^{t_2} \sum M_{ox} dt = (H_{ox})_2$$

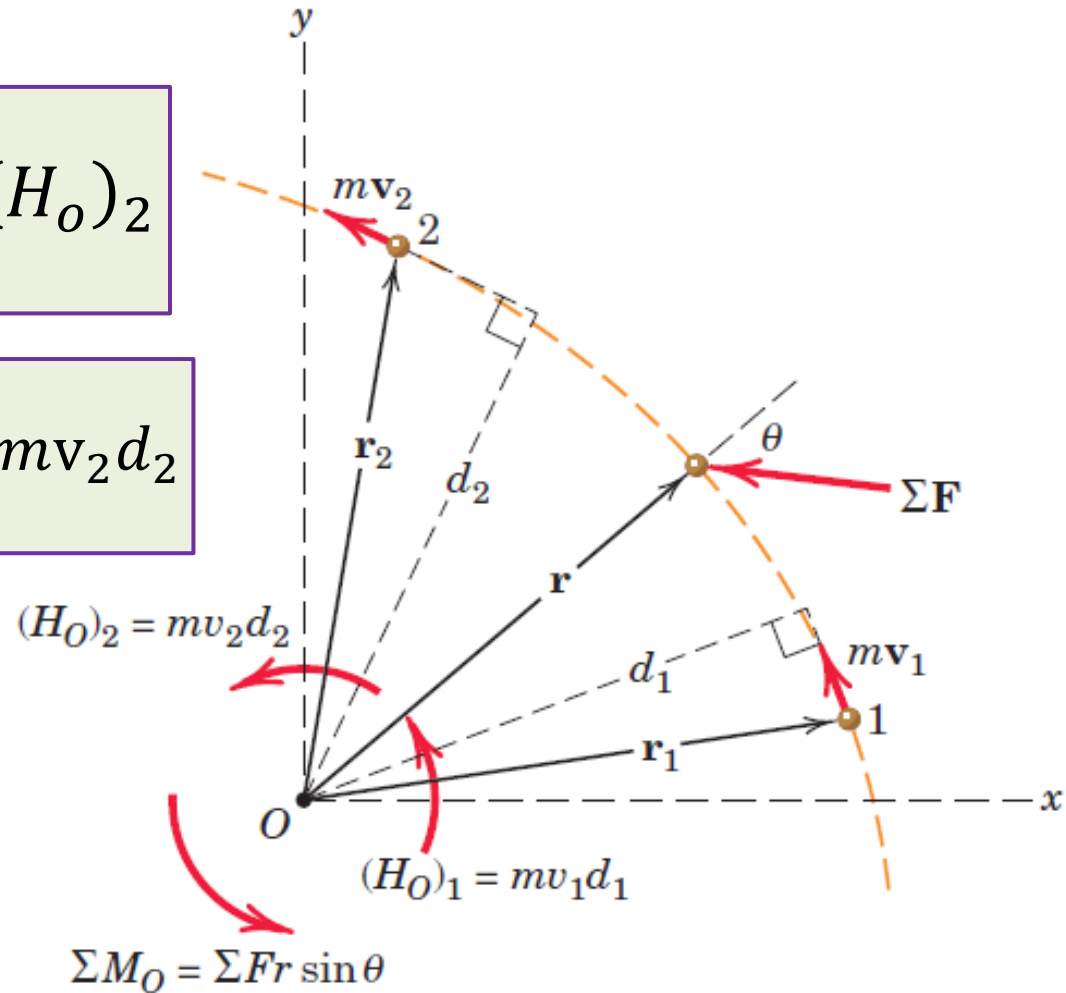
$$(H_{oy})_1 + \int_{t_1}^{t_2} \sum M_{oy} dt = (H_{oy})_2$$

$$(H_{oz})_1 + \int_{t_1}^{t_2} \sum M_{oz} dt = (H_{oz})_2$$

Plane Motion Application of Angular Impulse-Momentum Principle

$$(H_O)_1 + \int_{t_1}^{t_2} \sum M_O dt = (H_O)_2$$

$$mv_1 d_1 + \int_{t_1}^{t_2} \sum Fr \sin \theta dt = mv_2 d_2$$



Conservation of Angular Momentum

$$\int_{t_1}^{t_2} \sum M_o dt = \Delta H_o = (H_o)_2 - (H_o)_1$$

If **resultant moment** about a fixed point O of all forces acting on a particle is **zero** during an interval of time,

$$\Delta H_o = 0$$

$$\int_{t_1}^{t_2} \sum M_o dt = 0$$

$$(H_o)_1 = (H_o)_2$$

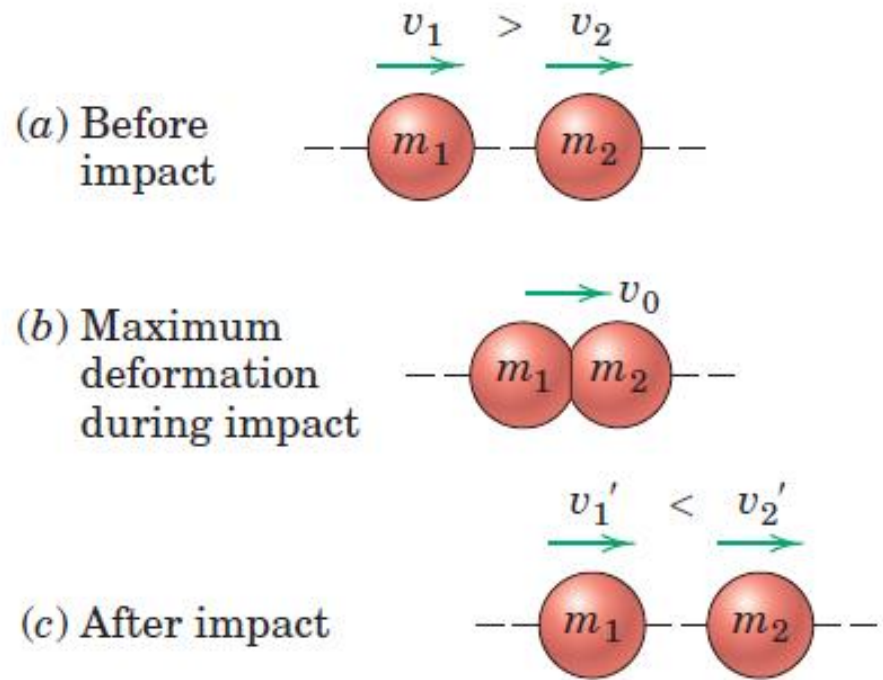
Resultant moment about a fixed point O of all forces acting on a particle is **zero** during an interval of time, its **angular momentum** H_o about that point remain **constant**

Impact

Impact refers to the collision between two bodies and is characterized by the generation of relatively large contact forces which act over a very short interval of time

Direct Central Impact

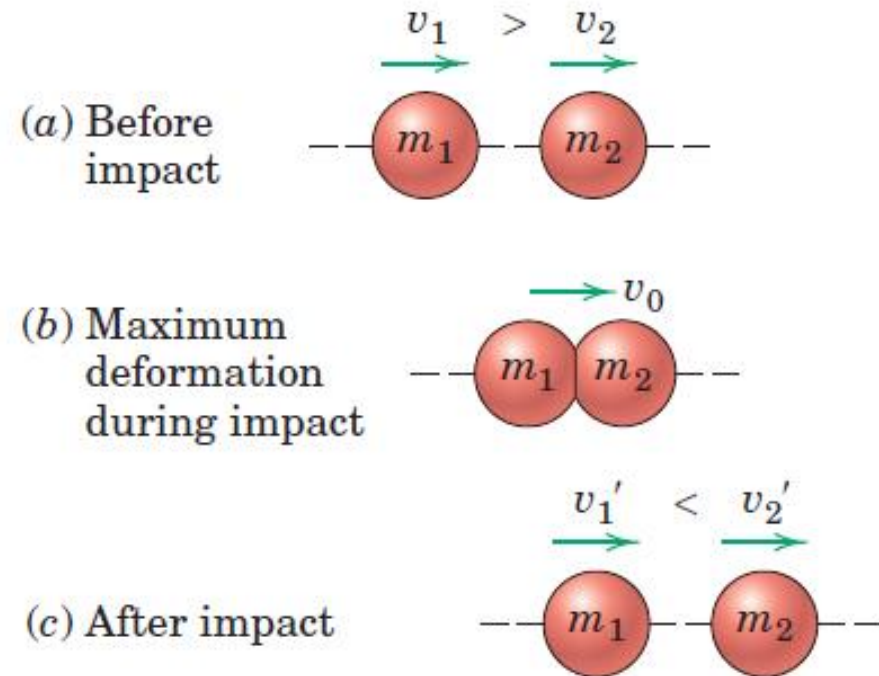
- Collinear motion of two spheres of masses m_1 and m_2
- v_1 is greater than v_2 , collision occurs with the contact forces directed along the line of centers
- Following initial contact, a short period of increasing deformation takes place until the contact area between the spheres ceases to increase.



Impact

Direct Central Impact

- Following initial contact, a short period of increasing deformation takes place until the contact area between the spheres ceases to increase. At this instant, both spheres, are moving with the same velocity v_0 .
- During the remainder of contact, a period of restoration occurs during which the contact area decreases to zero. In the final condition, the spheres now have new velocities v_1' , v_2' .



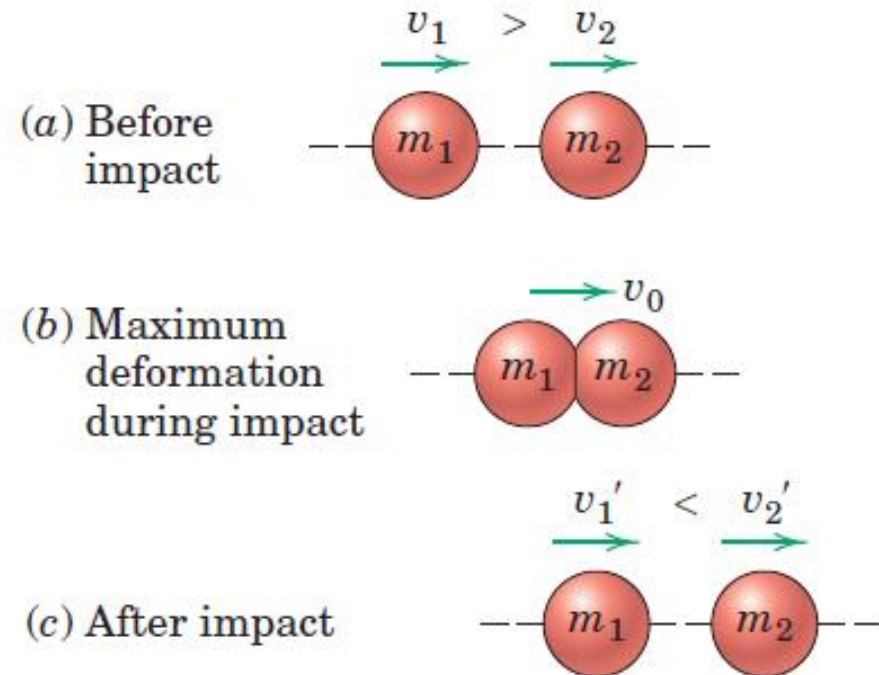
Impact

Direct Central Impact

- Because the contact forces are equal and opposite during impact, the linear momentum of the system remains unchanged

Law of conservation of linear momentum

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$



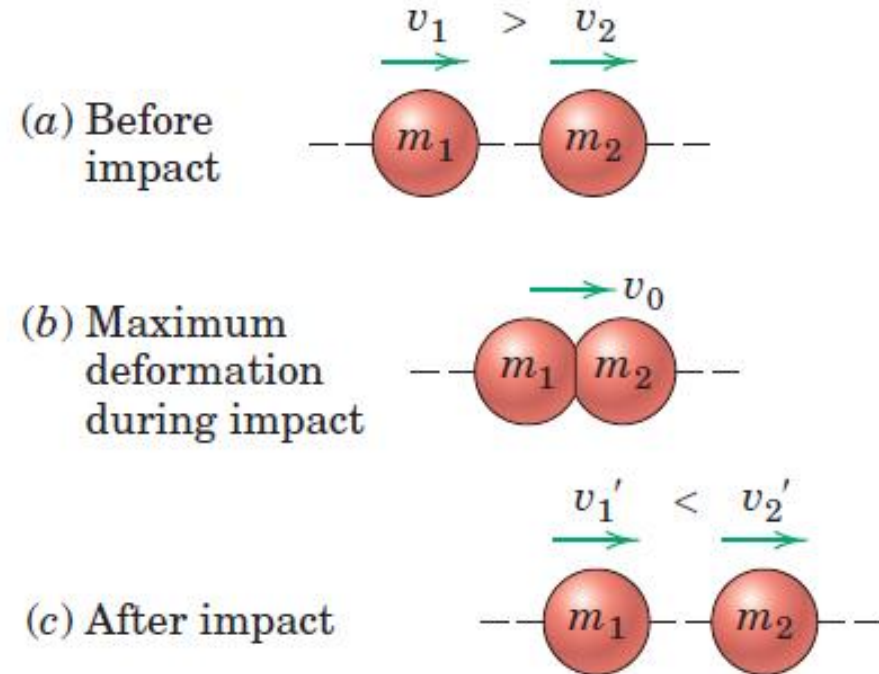
Coefficient of Restitution

Law of conservation of linear momentum

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$

Coefficient of restitution: Ratio e of the magnitude of the restoration impulse to the magnitude of the deformation impulse.

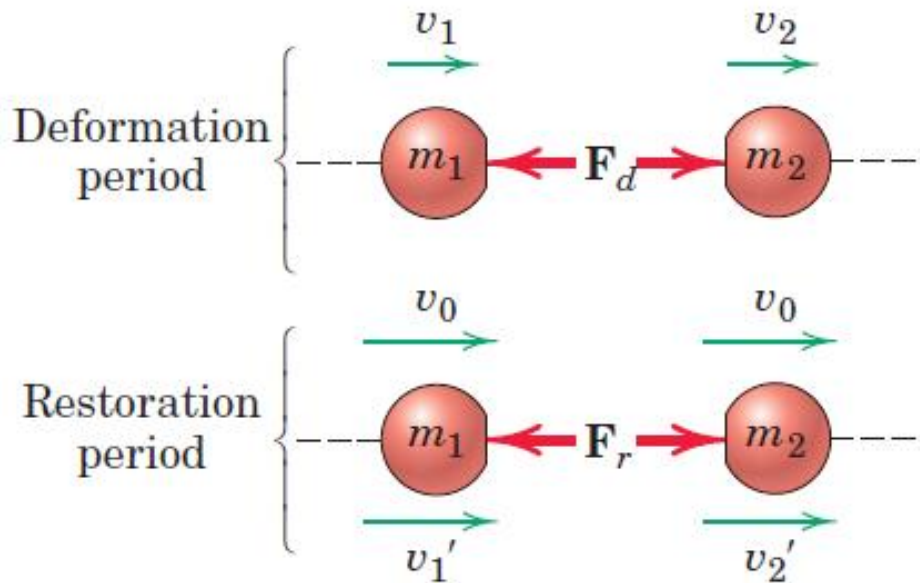
F_r and F_d are magnitudes of the contact forces during restoration and deformation periods.



Coefficient of Restitution

Coefficient of restitution: Ratio e of the magnitude of the restoration impulse to the magnitude of the deformation impulse.

F_r and F_d are magnitudes of the contact forces during restoration and deformation periods.



$$e = \frac{\int_{t_0}^t F_r dt}{\int_0^{t_0} F_d dt} = \frac{m_1[-v_1' - (-v_0)]}{m_1[-v_0 - (-v_1)]} = \frac{v_0 - v_1'}{v_1 - v_0}$$

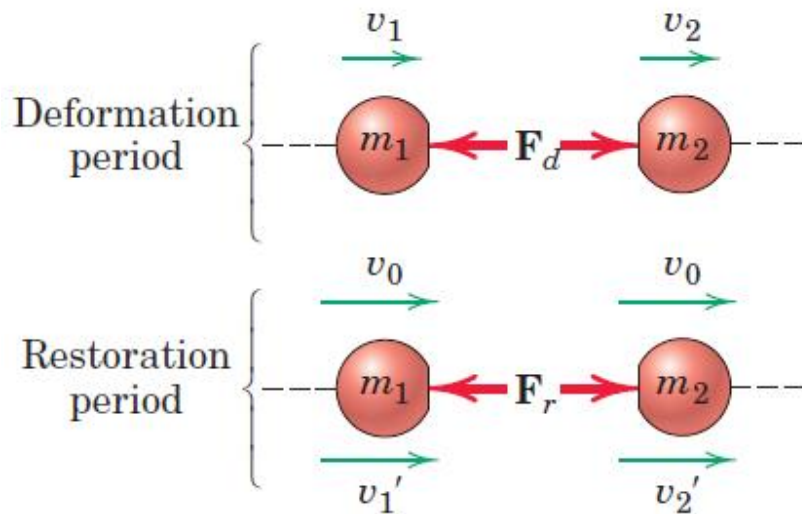
Coefficient of Restitution

For Particle 1

$$e = \frac{\int_{t_0}^t F_r dt}{\int_0^{t_0} F_d dt} = \frac{m_1[-v'_1 - (-v_0)]}{m_1[-v_0 - (-v_1)]} = \frac{v_0 - v'_1}{v_1 - v_0}$$

For Particle 2

$$e = \frac{\int_{t_0}^t F_r dt}{\int_0^{t_0} F_d dt} = \frac{m_2[v'_2 - (v_0)]}{m_2[v_0 - (v_2)]} = \frac{v'_2 - v_0}{v_0 - v_2}$$



Express the change of momentum
(and therefore v) in the same
direction as the impulse

Coefficient of Restitution

For Particle 1

$$e = \frac{\int_{t_0}^t F_r dt}{\int_0^{t_0} F_d dt} = \frac{m_1[-v_1' - (-v_0)]}{m_1[-v_0 - (-v_1)]} = \frac{v_0 - v_1'}{v_1 - v_0}$$

For Particle 2

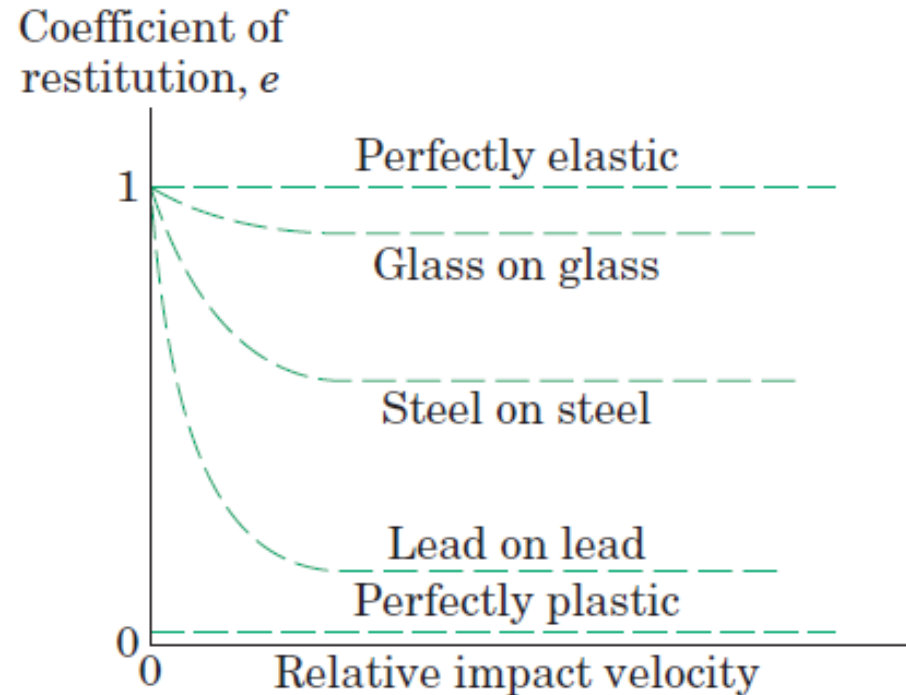
$$e = \frac{\int_{t_0}^t F_r dt}{\int_0^{t_0} F_d dt} = \frac{m_2[v_2' - (v_0)]}{m_2[v_0 - (v_2)]} = \frac{v_2' - v_0}{v_0 - v_2}$$

$$\frac{v_0 - v_1'}{v_1 - v_0} = \frac{v_2' - v_0}{v_0 - v_2}$$

$$e = \frac{v_2' - v_1'}{v_1 - v_2} = \frac{|relative\ velocity\ of\ separation|}{|relative\ velocity\ of\ approach|}$$

Energy Loss During Impact

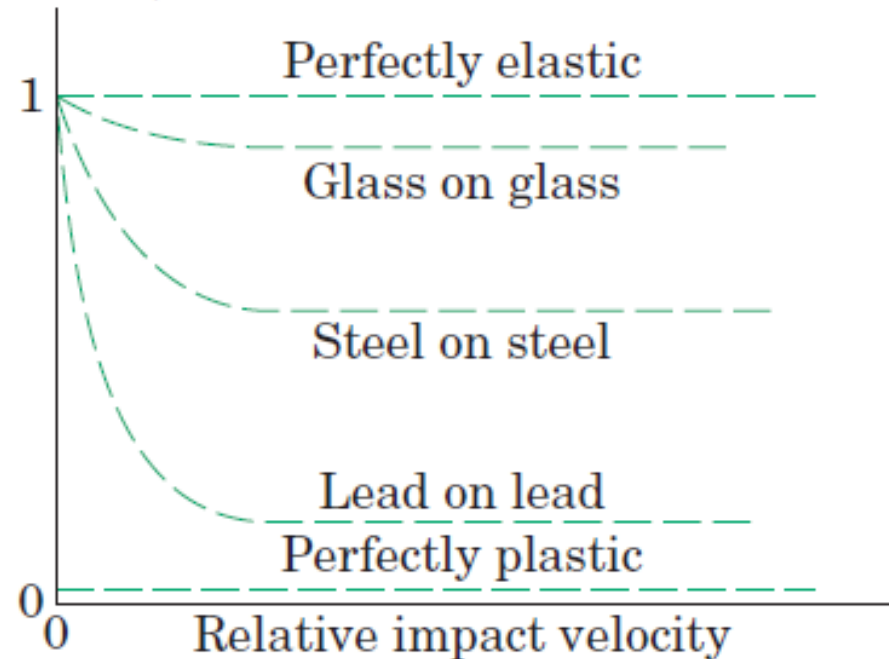
- Energy loss = KE before impact - KE after impact
- Value $e=1$ means that the capacity of the two particles to recover equals their tendency to deform. This condition is one of *elastic impact* with no energy loss.
- Value $e=0$ describes *inelastic* or *plastic impact* where the particles cling together after collision and the loss of energy is a maximum.
- All impact conditions lie somewhere between these two extremes.



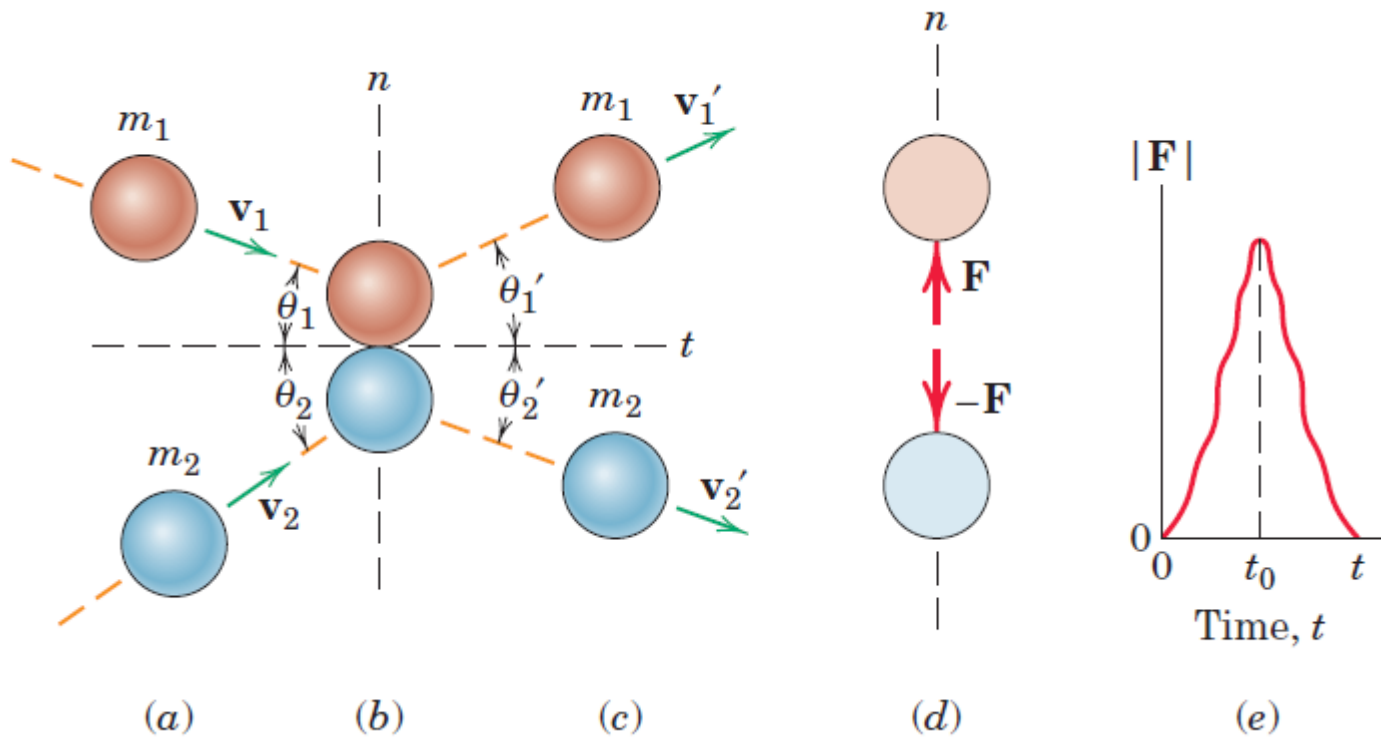
Energy Loss During Impact

- Coefficient of restitution must be associated with a pair of contacting bodies.
- Coefficient of restitution depends upon given geometries and a given combination of contacting materials, and also on the impact velocity
- It approaches unity as the impact velocity approaches zero

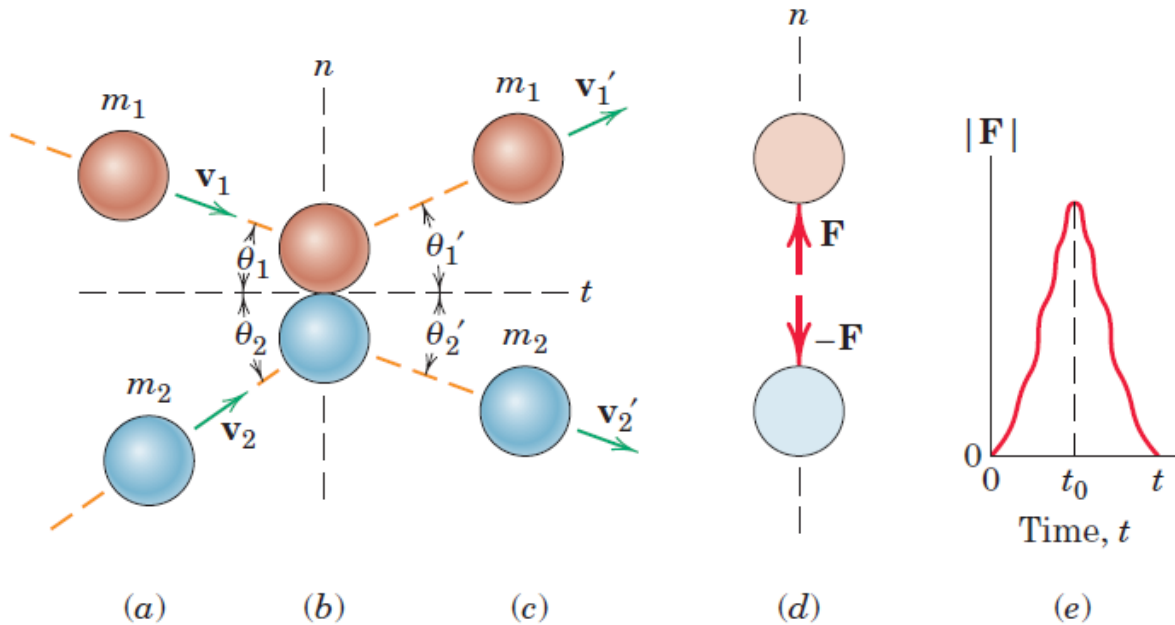
Coefficient of restitution, e



Oblique Central Impact

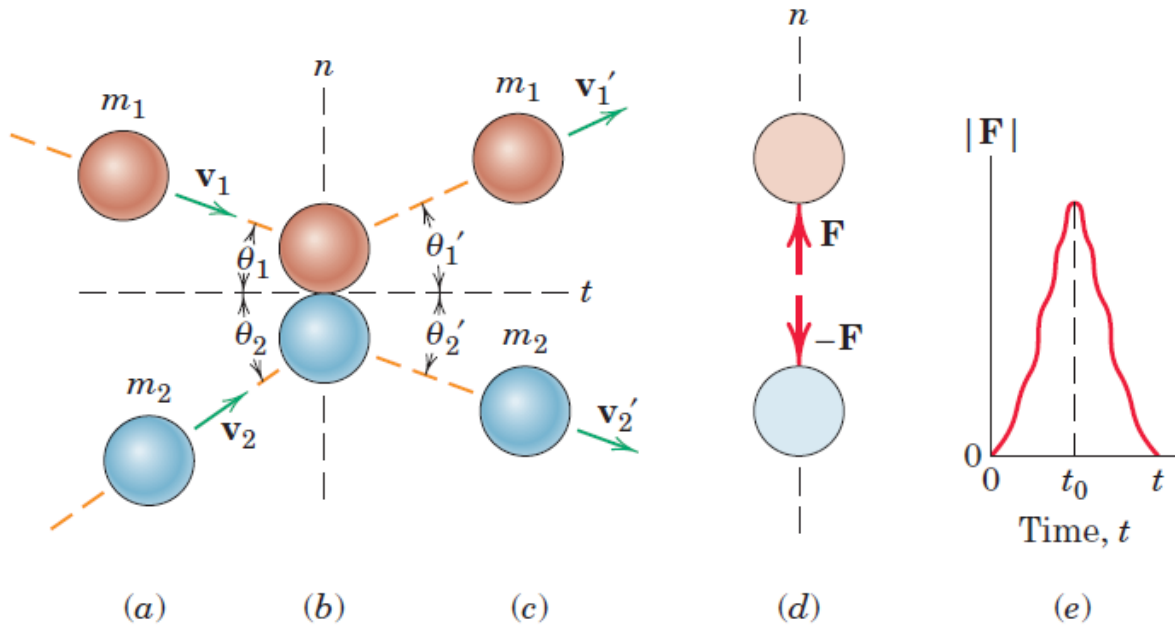


Oblique Central Impact



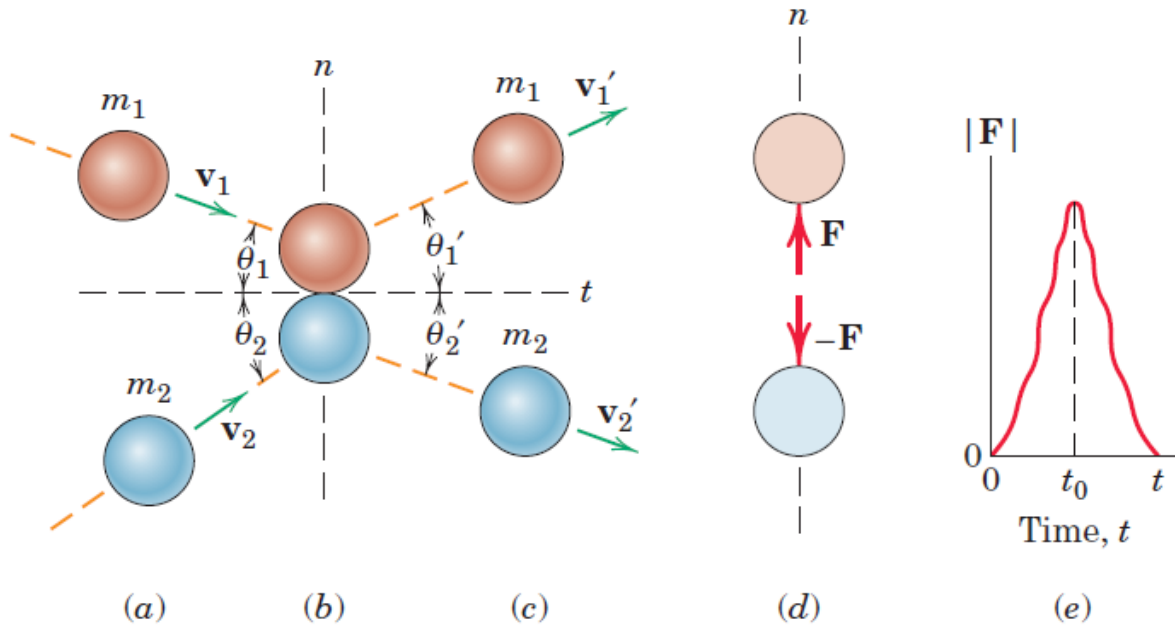
- Spherical particles of mass m_1 and m_2 have initial velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ in the same plane and approach each other on a collision course. The directions of the velocity vectors are measured from the direction tangent to the contacting surfaces

Oblique Central Impact



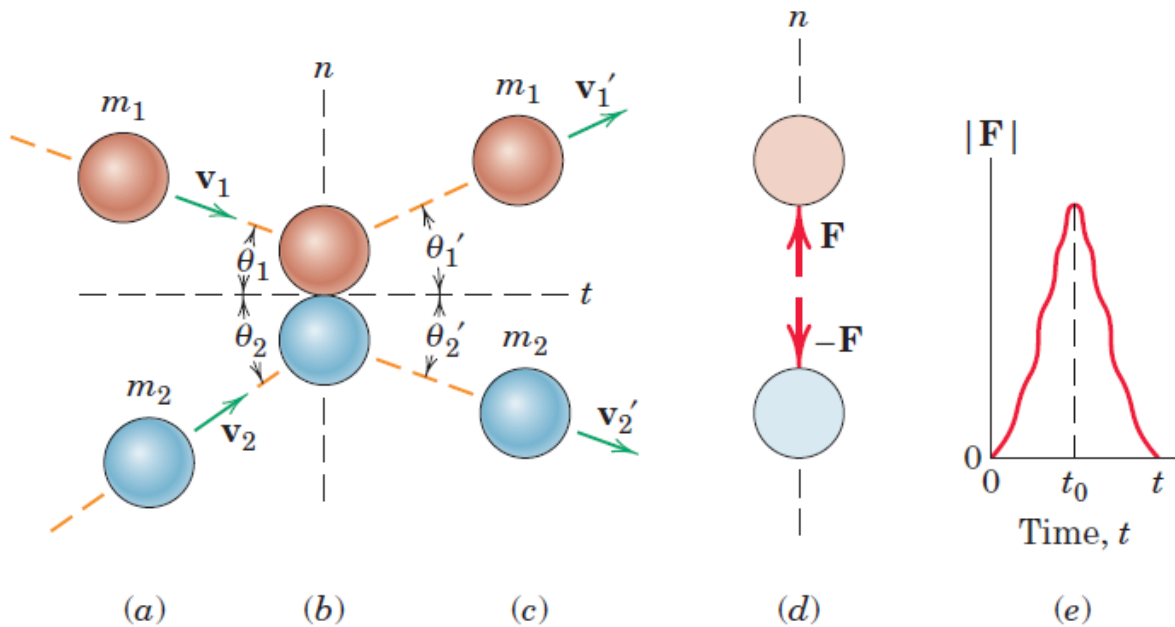
- The impact forces are $\mathbf{F}$ and $-\mathbf{F}$
- They vary from zero to their peak value during the deformation portion of the impact and back again to zero during the restoration period

Oblique Central Impact



- Momentum of the system is conserved in the n -direction
- The momentum for each particle is conserved in the t -direction since there is no impulse on either particle in the t -direction
- The coefficient of restitution, as in the case of direct central impact, is the positive ratio of the recovery impulse to the deformation impulse

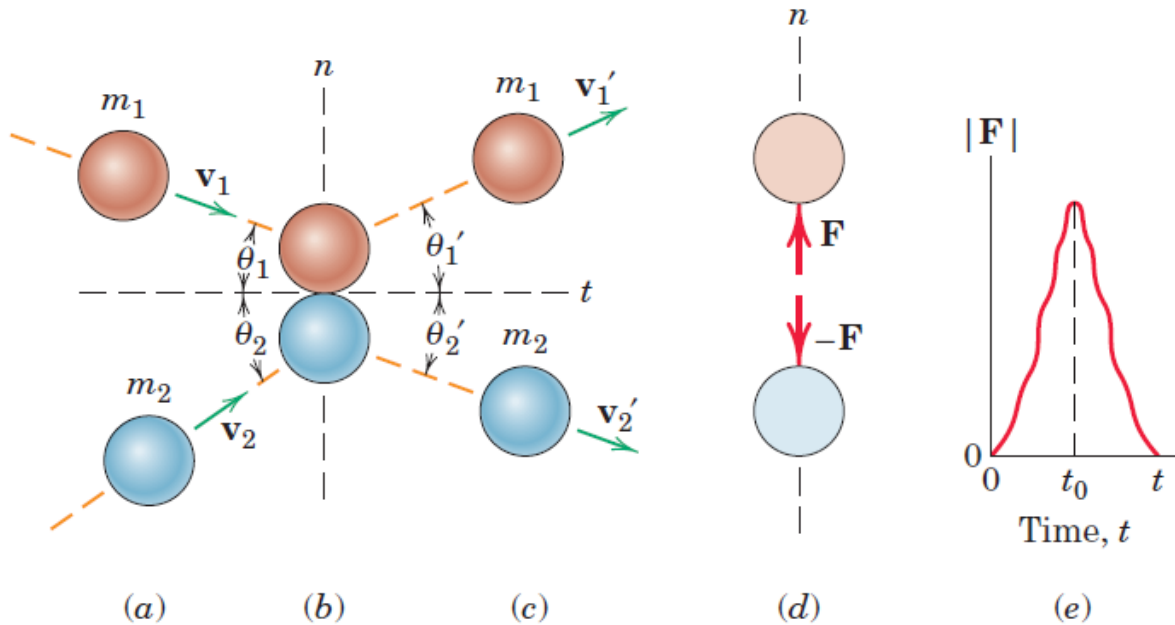
Oblique Central Impact



➤ Momentum of the system is conserved in the n -direction

$$m_1 v_{1n} + m_2 v_{2n} = m_1 v'_{1n} + m_2 v'_{2n}$$

Oblique Central Impact

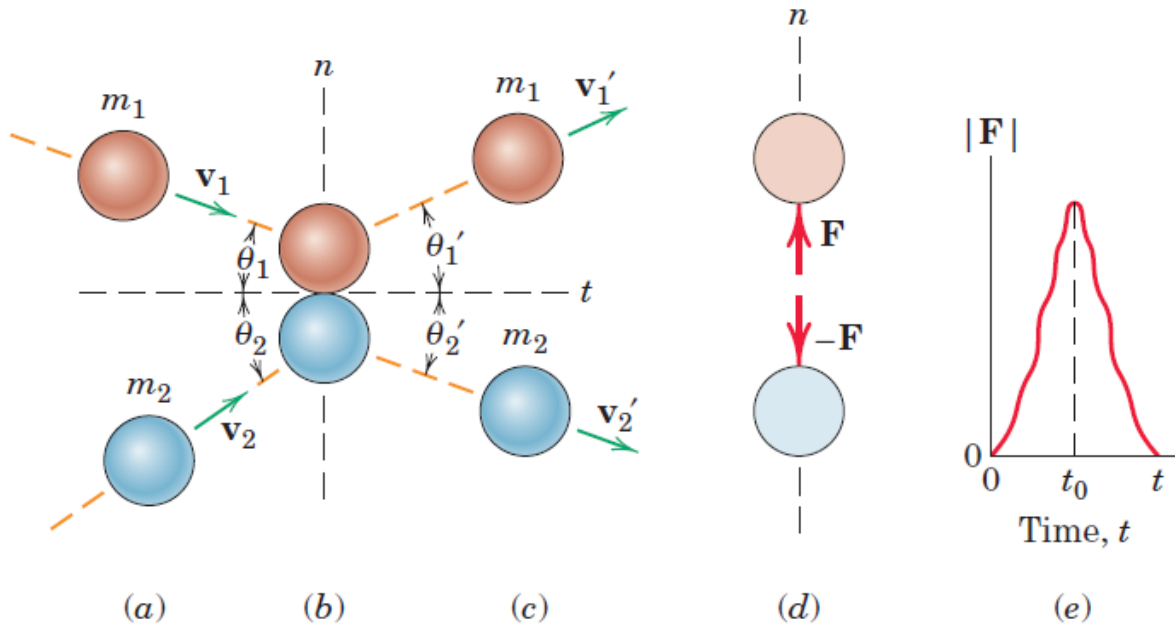


- The momentum for each particle is conserved in the t -direction since there is no impulse on either particle in the t -direction

$$m_1 v_{1t} = m_1 v'_{1t}$$

$$m_2 v_{2t} = m_2 v'_{2t}$$

Oblique Central Impact



- The coefficient of restitution, as in the case of direct central impact, is the positive ratio of the recovery impulse to the deformation impulse

$$e = \frac{v_{2n}' - v_{1n}'}{v_{1n} - v_{2n}}$$

Relative Motion

x-y-z are parallel to X-Y-Z

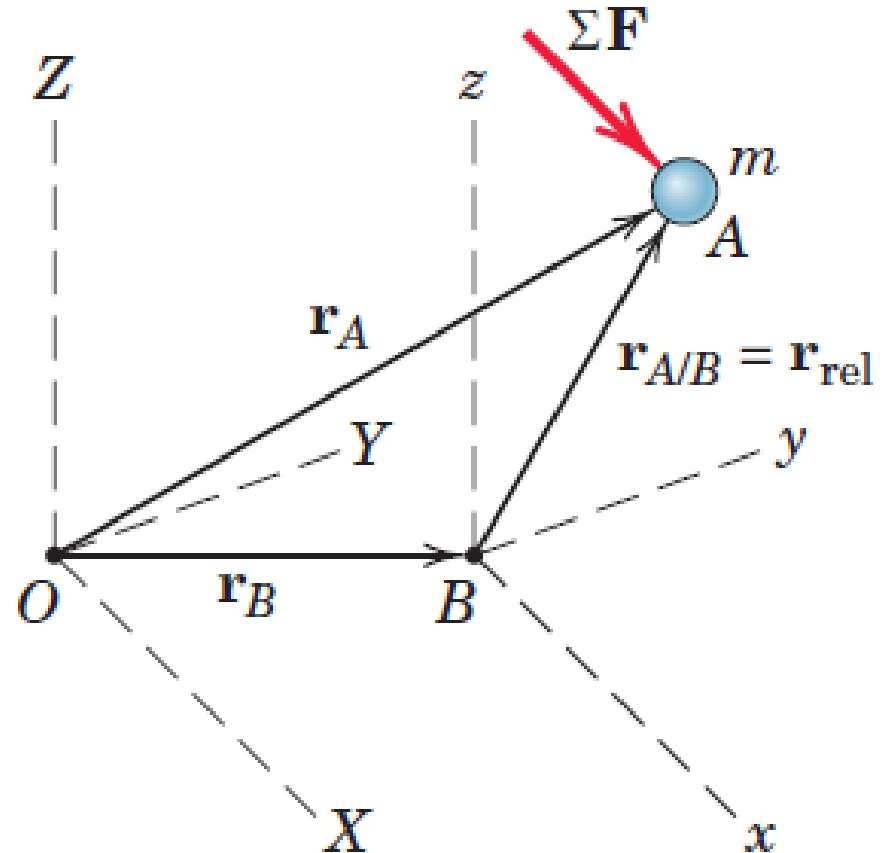
Point B is accelerating with a_B

$$a_{A/B} = \ddot{r}_{A/B}$$

$$a_A = a_B + a_{A/B}$$

Newton's second law of motion

$$\sum F = ma_A$$



D'Alembert's Principle

Newton's second law of motion

$$\sum F = ma_A$$

- The system x-y-z attached to the particle, the particle necessarily appears to be at rest or in equilibrium in x-y-z
- The observer who is accelerating with x-y-z concludes that a force $-m\mathbf{a}$ acts on the particle to balance $\sum \mathbf{F}$

$$\sum F - ma_A = 0$$

This fictitious force is known as the *inertia force*, and the artificial state of equilibrium created is known as *dynamic equilibrium*.

The apparent transformation of a problem in dynamics to one in statics has become known as *D'Alembert's principle*

D'Alembert's Principle

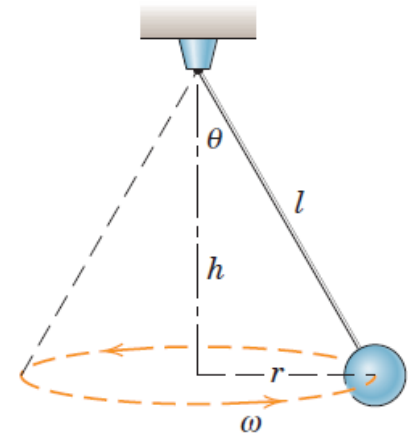
$$\sum F = ma_n$$

$$T \sin \theta = mr\omega^2$$

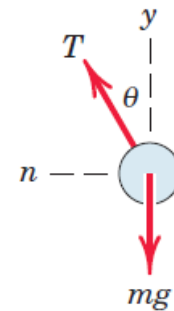
$$T \cos \theta = mg$$

Adding $mr\omega^2$ opposite to the direction of acceleration

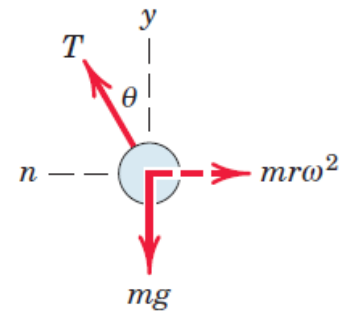
$$T \sin \theta - mr\omega^2 = 0$$



(a)



(b)

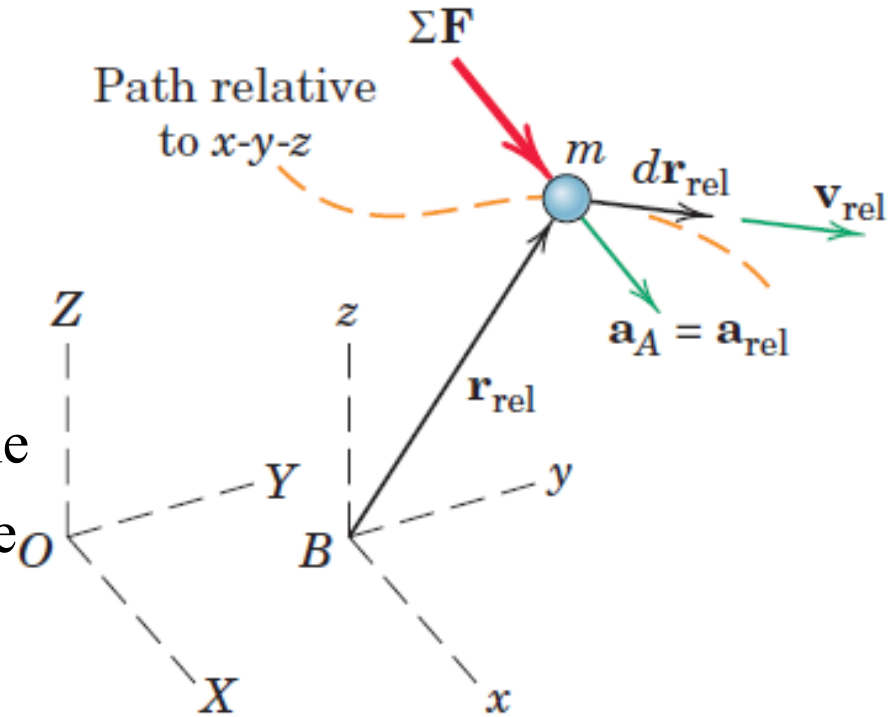


(c)

Constant velocity Non-rotating system

$$\sum \mathbf{F} = m\mathbf{a} = m\mathbf{a}_{rel}$$

Are the work-energy equation and the impulse-momentum equation relative to a constant-velocity, nonrotating system valid?



Work-Energy Equation in Constant velocity

Non-rotating system

Work done by $\sum F$ relative to x - y - z

$$dU_{rel} = \sum \mathbf{F} \cdot d\mathbf{r}_{rel}$$

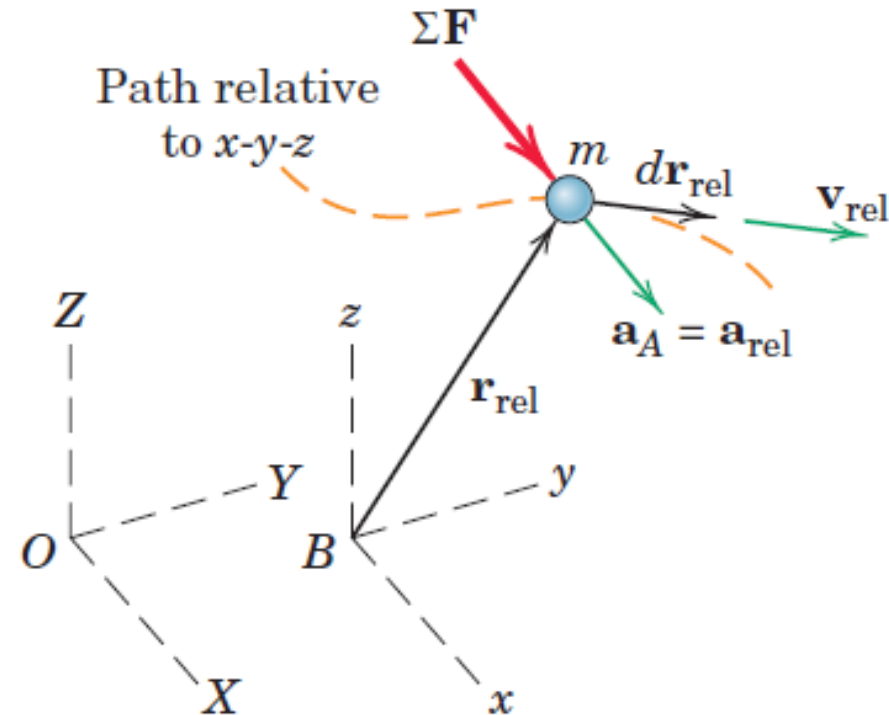
$$= m\mathbf{a} \cdot d\mathbf{r}_{rel} = m\mathbf{a}_{rel} \cdot d\mathbf{r}_{rel}$$

$$\mathbf{a}_{rel} \cdot d\mathbf{r}_{rel} = \mathbf{v}_{rel} \cdot d\mathbf{v}_{rel}$$

$$dU_{rel} = m\mathbf{a}_{rel} \cdot d\mathbf{r}_{rel}$$

$$= m\mathbf{v}_{rel} \cdot d\mathbf{v}_{rel} = mv_{rel}dv_{rel}$$

$$= d\left(\frac{1}{2}mv_{rel}^2\right) = dT_{rel}$$



Work-Energy Equation in Constant velocity

Non-rotating system

Work done by $\sum F$ relative to x - y - z

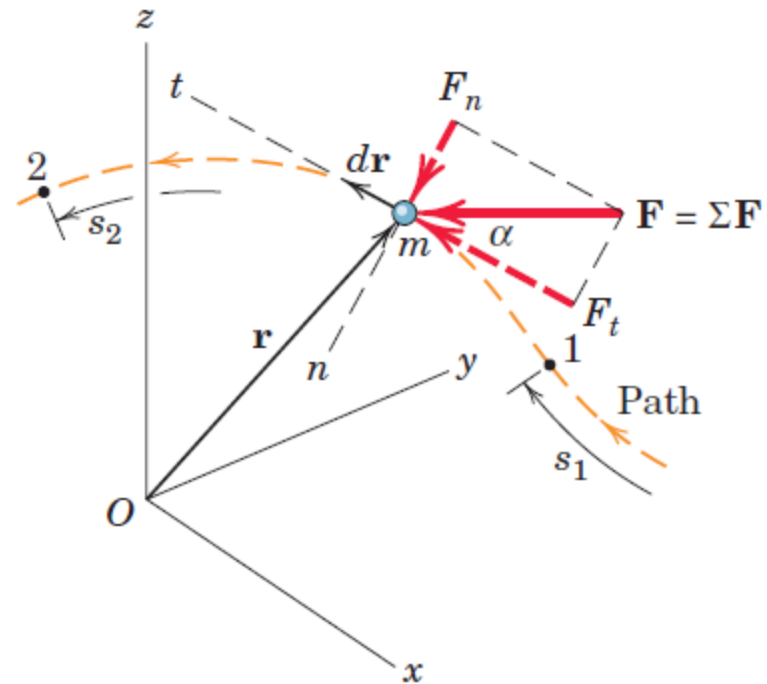
$$dU_{rel} = m \mathbf{a}_{rel} \cdot d\mathbf{r}_{rel} = m \mathbf{v}_{rel} \cdot d\mathbf{v}_{rel} = m v_{rel} dv_{rel}$$

$$= d \left(\frac{1}{2} m v_{rel}^2 \right) = dT_{rel}$$

$$\mathbf{a} \cdot d\mathbf{r} = a_t ds$$

$$a_t ds = v dv$$

$$a_{t_rel} ds_{rel} = v_{rel} dv_{rel}$$



Work-Energy Equation in Constant velocity

Non-rotating system

Work done by $\sum F$ relative to x - y - z

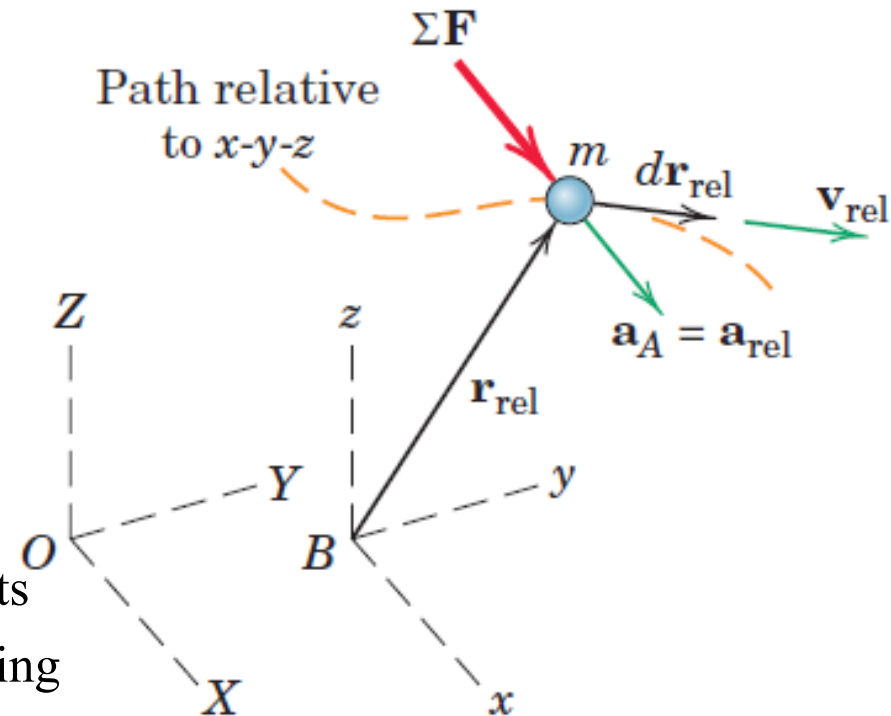
$$dU_{rel} = m \mathbf{a}_{rel} \cdot d\mathbf{r}_{rel} = m \mathbf{v}_{rel} \cdot d\mathbf{v}_{rel} = m v_{rel} dv_{rel}$$

$$= d \left(\frac{1}{2} m v_{rel}^2 \right) = dT_{rel}$$

$$dU_{rel} = dT_{rel}$$

$$U_{rel} = \Delta T_{rel}$$

Work-energy equation holds for measurements made relative to a constant-velocity, nonrotating system



Impulse-momentum equation in Constant velocity Non-rotating system

$$\sum \mathbf{F} = m\mathbf{a} = m\mathbf{a}_{rel} \qquad \sum \mathbf{F} dt = m\mathbf{a}dt = m\mathbf{a}_{rel}dt$$

$$\sum \mathbf{F} dt = m\mathbf{a}_{rel}dt = m d\mathbf{v}_{rel} = d(m\mathbf{v}_{rel})$$

$$\sum \mathbf{F} dt = d(m\mathbf{v}_{rel}) = d(\mathbf{G}_{rel})$$

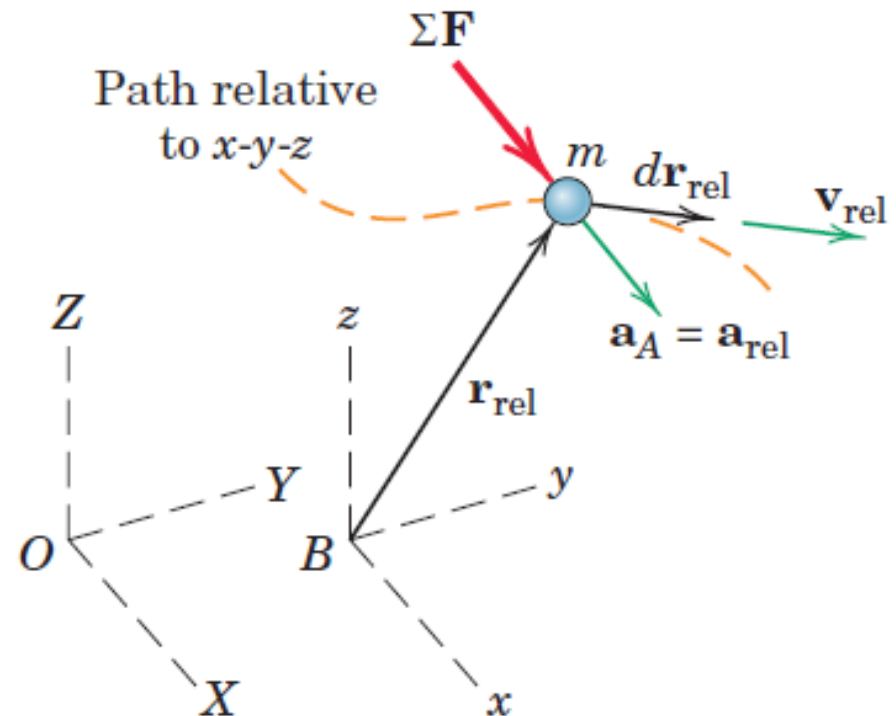
$$\sum \mathbf{F} = (\dot{\mathbf{G}}_{rel}) \qquad \int \sum \mathbf{F} dt = \Delta \mathbf{G}_{rel}$$

Impulse-momentum equations for a fixed reference system also hold for measurements made relative to a constant-velocity, nonrotating system

Moment-Angular Momentum Equation in Constant velocity Non-rotating system

Relative angular momentum of the particle about a point in x - y - z , such as the origin B , as the moment of the relative linear momentum

$$(\mathbf{H}_B)_{rel} = \mathbf{r}_{rel} \times \mathbf{G}_{rel}$$



Moment-Angular Momentum Equation in Constant velocity Non-rotating system

Relative angular momentum of the particle about a point in x-y-z, such as the origin B , as the moment of the relative linear momentum

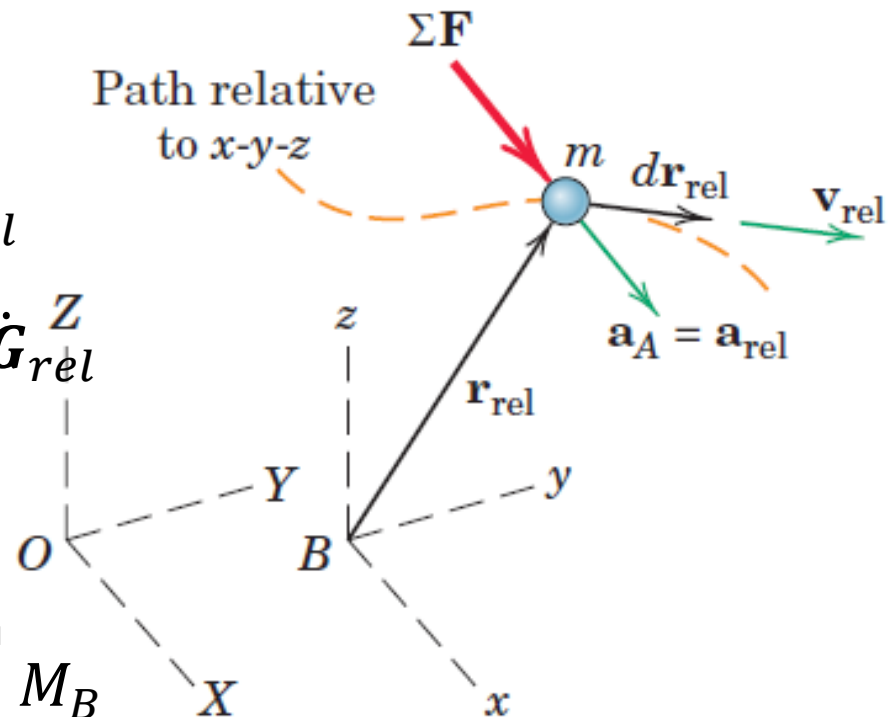
$$(\mathbf{H}_B)_{rel} = \mathbf{r}_{rel} \times \mathbf{G}_{rel}$$

$$(\dot{\mathbf{H}}_B)_{rel} = \dot{\mathbf{r}}_{rel} \times \mathbf{G}_{rel} + \mathbf{r}_{rel} \times \dot{\mathbf{G}}_{rel}$$

$$(\dot{\mathbf{H}}_B)_{rel} = \mathbf{v}_{rel} \times m\mathbf{v}_{rel} + \mathbf{r}_{rel} \times \dot{\mathbf{G}}_{rel}$$

$$(\dot{\mathbf{H}}_B)_{rel} = 0 + \mathbf{r}_{rel} \times \dot{\mathbf{G}}_{rel}$$

$$(\dot{\mathbf{H}}_B)_{rel} = 0 + \mathbf{r}_{rel} \times \sum \mathbf{F} = \sum \mathbf{M}_B$$

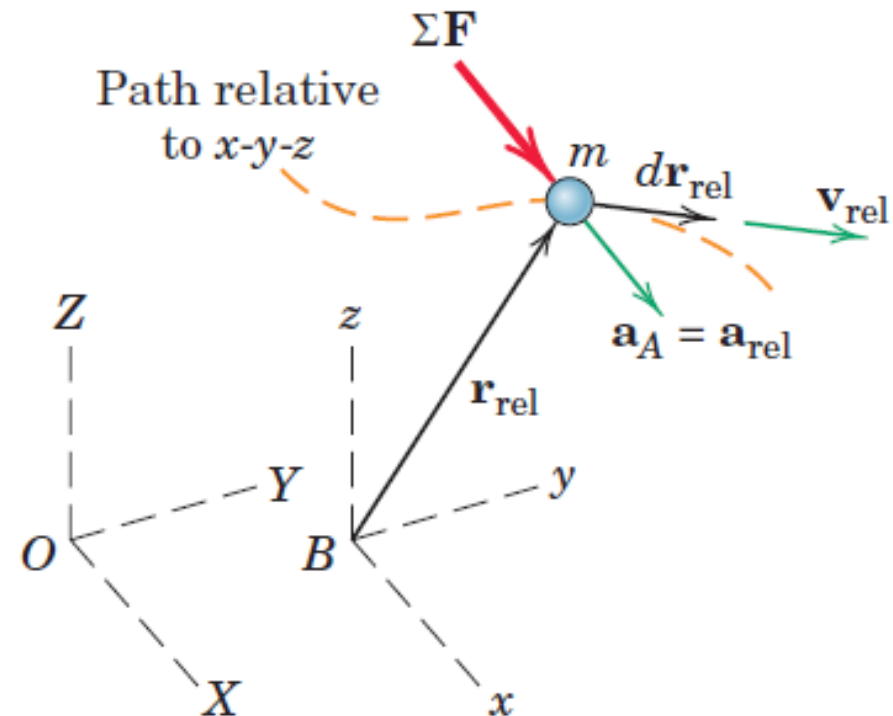


Moment-Angular Momentum Equation in Constant velocity Non-rotating system

Rate of change of angular momentum about point B is equal to sum of the moments about B of all forces on m

$$(\dot{\mathbf{H}}_B)_{rel} = \sum M_B$$

Moment-angular momentum relation holds with respect to a constant-velocity, nonrotating system



Difference between Absolute and Relative Quantities

Although the work-energy and impulse-momentum equations hold relative to a system translating with a constant velocity, the individual expressions for work, kinetic energy, and momentum differ between the fixed and the moving systems

$$\mathbf{G} = m\mathbf{v}_A \quad \neq \quad \mathbf{G}_{rel} = m\mathbf{v}_{rel}$$

$$dU = \sum \mathbf{F} \cdot d\mathbf{r} \quad \neq \quad dU_{rel} = \sum \mathbf{F} \cdot d\mathbf{r}_{rel}$$

$$T = \frac{1}{2}mv^2 \quad \neq \quad T_{rel} = \frac{1}{2}mv_{rel}^2$$

Central Force Motion

When a particle **moves under the influence of a force directed toward a fixed center of attraction**, the motion is called CENTRAL FORCE MOTION

Examples: Orbital movement of planets and satellites

The laws which govern this motion were deduced from observation of the motions of the planets by J. Kepler (1571–1630).

An understanding of central-force motion is required to design high-altitude rockets, earth satellites, and space vehicles

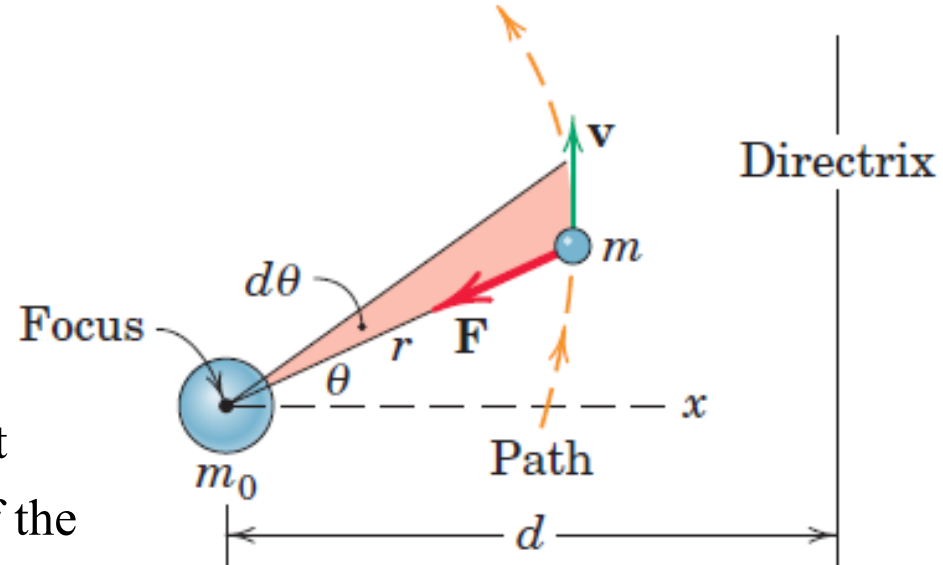
Central Force Motion

$$F = G \frac{mm_0}{r^2}$$

m_0 is the mass of the attracting body
and it is assumed to be fixed

G is the universal gravitational constant

r is the distance between the centers of the
masses



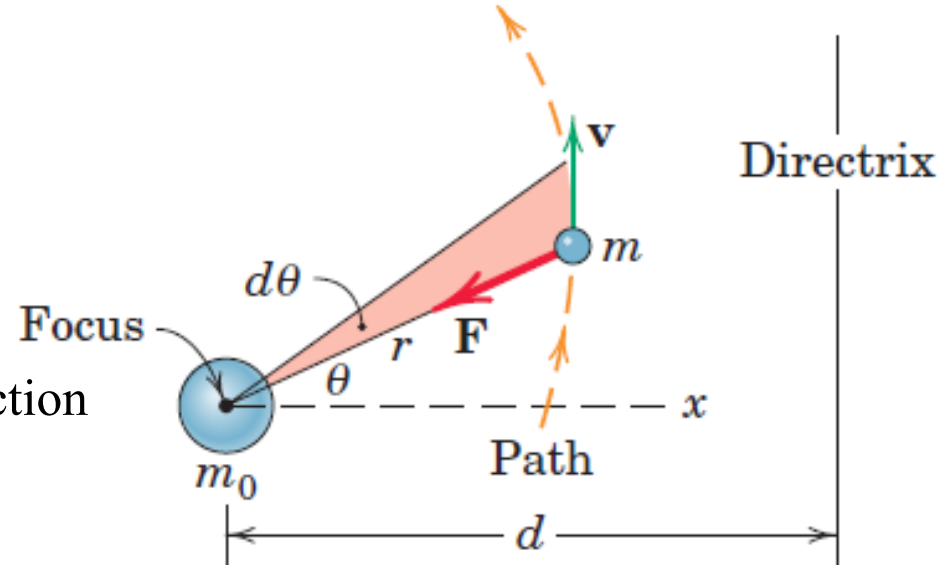
The particle of mass m could represent the **earth moving about the sun**, the **moon moving about the earth**, or a **satellite in its orbital motion about the earth above the atmosphere**

Central Force Motion

$$F = G \frac{mm_0}{r^2}$$

In polar coordinates (r - θ coordinates),

- F will always be in the negative r -direction
- There is no force in the θ -direction



$$-G \frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$

$$0 = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

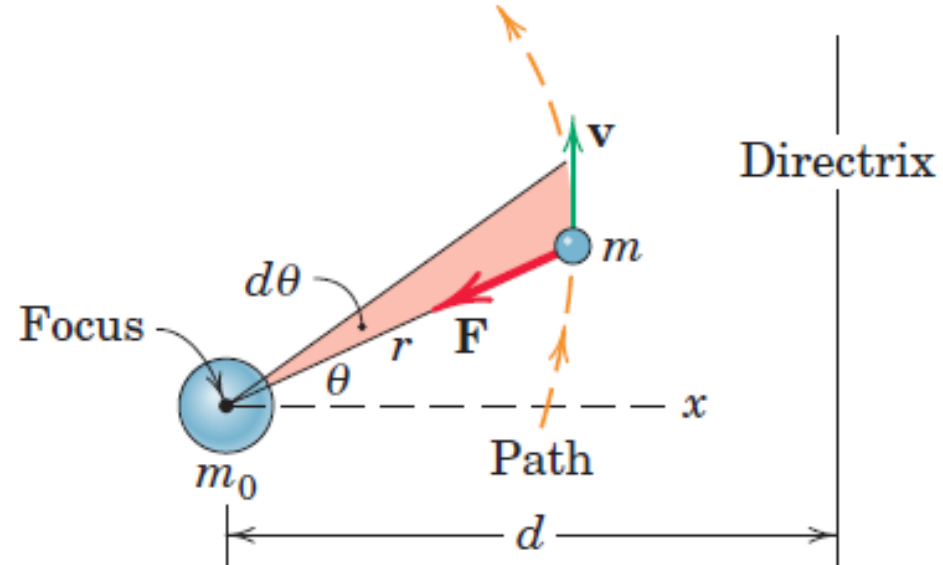
Central Force Motion

$$-G \frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$

$$0 = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

$$0 = \frac{d(r^2\dot{\theta})}{dt} = 2r\dot{r}\dot{\theta} + r^2\ddot{\theta}$$

$$r^2\dot{\theta} = h = \text{constant}$$



Central Force Motion

$$r^2 \dot{\theta} = h = \text{constant}$$

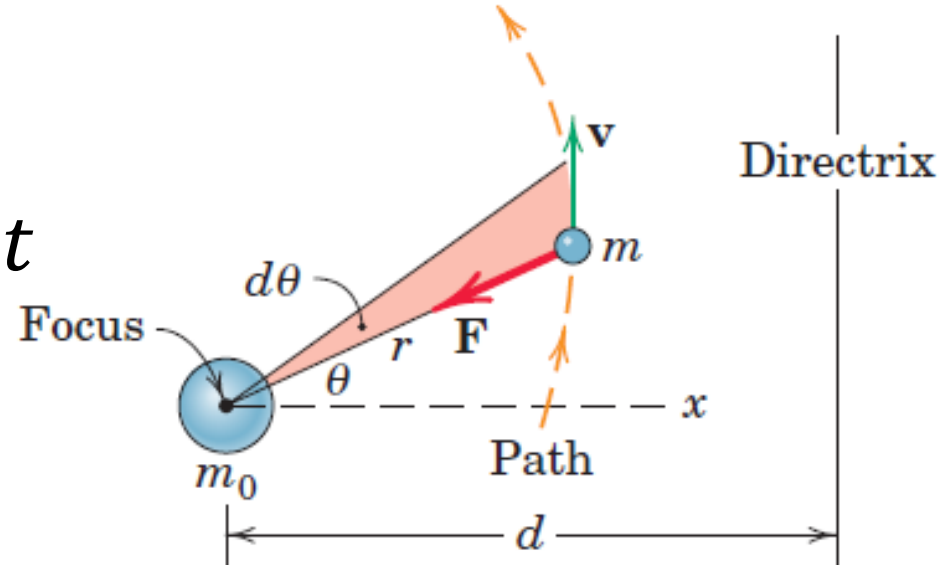
$$mr^2 \dot{\theta} = h_1 = \text{constant}$$

$$\mathbf{v} = \dot{r} \mathbf{e}_r + r \dot{\theta} \mathbf{e}_\theta$$

$$\mathbf{H} = \mathbf{r} \times m \mathbf{v}$$

$$|\mathbf{H}| = |\mathbf{r} \times m \mathbf{v}| = mr^2 \dot{\theta} = h_1 = \text{constant}$$

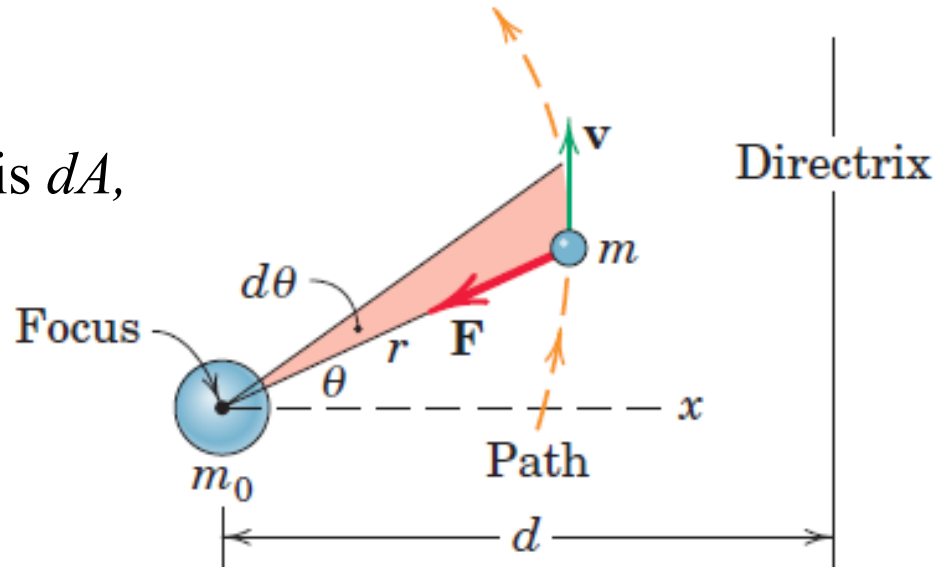
Angular momentum $\mathbf{H}$ remains constant (is conserved) if there is no moment acting on the particle about a fixed point O .



Central Force Motion

In time dt , area swept by mass m is dA ,

$$dA = \frac{1}{2} r (r d\theta) = \frac{1}{2} r^2 d\theta$$



$$\frac{dA}{dt} = \dot{A} = \frac{1}{2} r^2 \frac{d\theta}{dt} = \frac{1}{2} r^2 \dot{\theta} = \text{constant}$$

Kepler's *second law* of planetary motion, which states that the areas swept through in equal times are equal.

Central Force Motion

$$-G \frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$

$$r = \frac{1}{u} \quad \dot{r} = -\frac{1}{u^2} \dot{u}$$

$$\dot{r} = -r^2 \dot{u}$$

$$\dot{r} = -h \frac{\dot{u}}{\dot{\theta}}$$

$$r^2 \dot{\theta} = h = \text{constant}$$

$$\dot{r} = -h \frac{du}{d\theta}$$

$$\dot{r} = -h \frac{du}{d\theta}$$

$$\ddot{r} = -h \frac{d^2 u}{d\theta^2} \dot{\theta}$$

$$\ddot{r} = -\frac{h^2}{r^2} \frac{d^2 u}{d\theta^2}$$

$$\ddot{r} = -h^2 u^2 \frac{d^2 u}{d\theta^2}$$

Using expression for $\ddot{r}$, r ,
 $r^2 \dot{\theta} = h$

$$-Gm_0 u^2 = -h^2 u^2 \frac{d^2 u}{d\theta^2} - \frac{1}{u} h^2 u^4$$

Central Force Motion

$$-G \frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$

$$-Gm_0u^2 = -h^2u^2 \frac{d^2u}{d\theta^2} - \frac{1}{u}h^2u^4$$

$$\frac{d^2u}{d\theta^2} + u = \frac{Gm_0}{h^2}$$

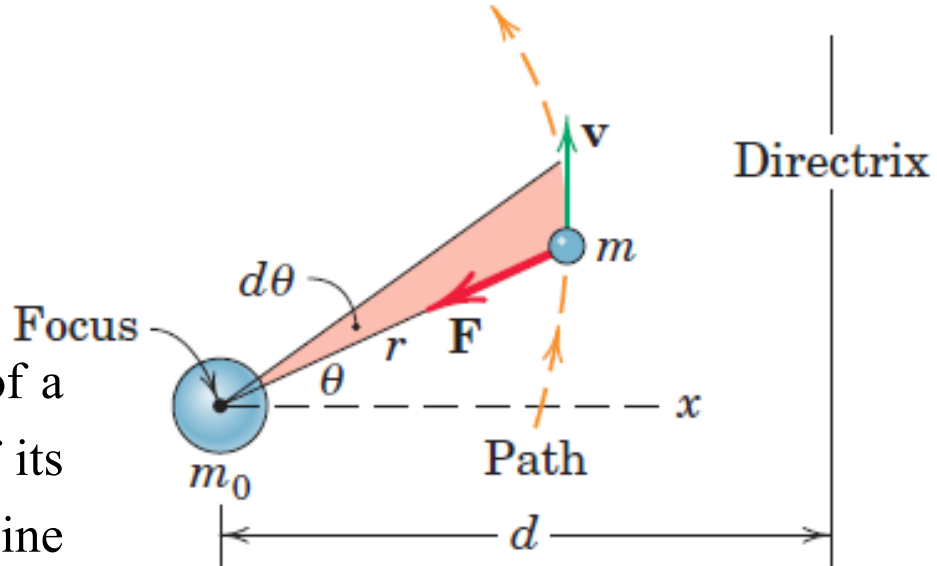
$$u = \frac{1}{r} = C \cos(\theta + \delta) + \frac{Gm_0}{h^2}$$

$$u = C \cos(\theta) + \frac{Gm_0}{h^2} \quad \delta = 0 \text{ if x-axis is chosen such that } r \text{ is minimum when } \theta \text{ is zero.}$$

Central Force Motion

$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

A conic section is formed by the locus of a point which moves so that the ratio e of its distance from a point (focus) to a line (directrix) is constant



$$e = \frac{r}{d - r \cos \theta}$$

$$\frac{ed - e r \cos \theta}{r} = \frac{ed}{r} - e \cos \theta = 1$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos \theta}{d}$$

$$\frac{1}{r} = \frac{e \cos \theta + 1}{ed}$$

Central Force Motion

$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

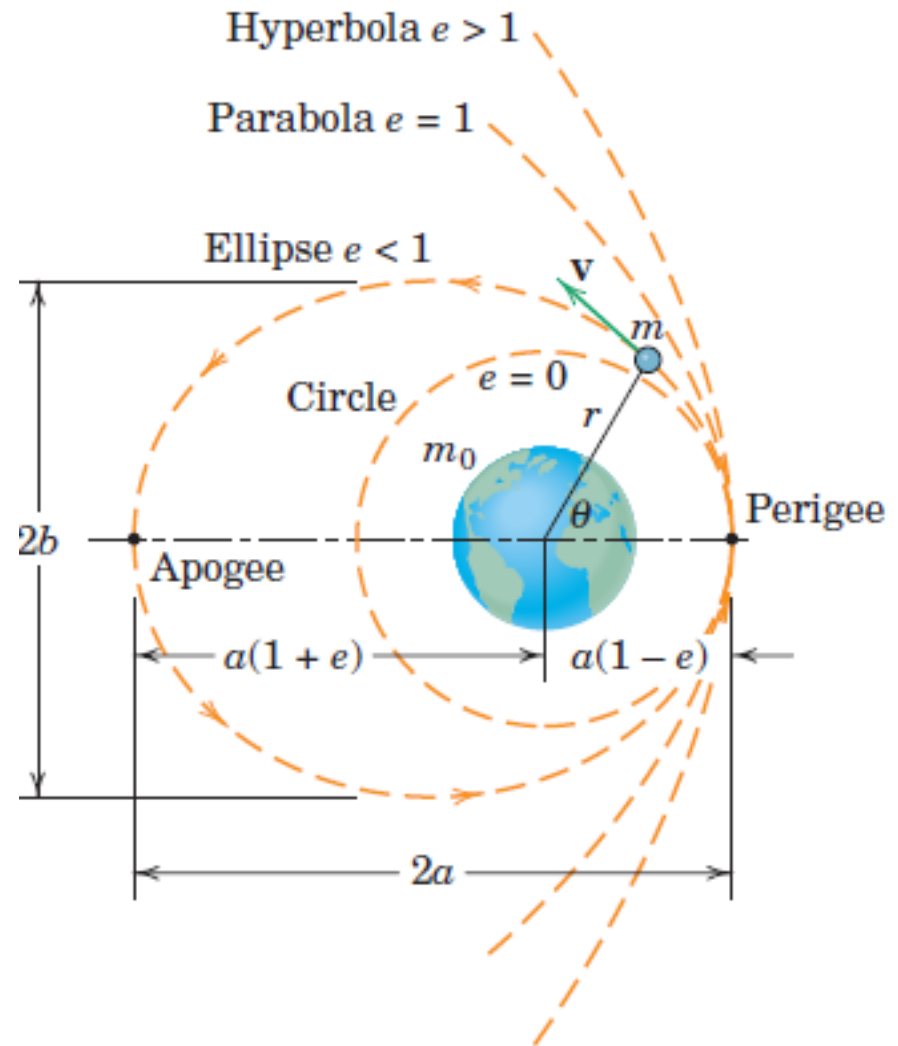
$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d}$$

$$C = \frac{1}{d} \quad ed = \frac{h^2}{Gm_0}$$

$$C = \frac{1}{d} \quad e = \frac{h^2 C}{Gm_0}$$

$e < 1$ Ellipse $e = 1$ Parabola

$e > 1$ Hyperbola



Central Force Motion Ellipse, $e < 1$

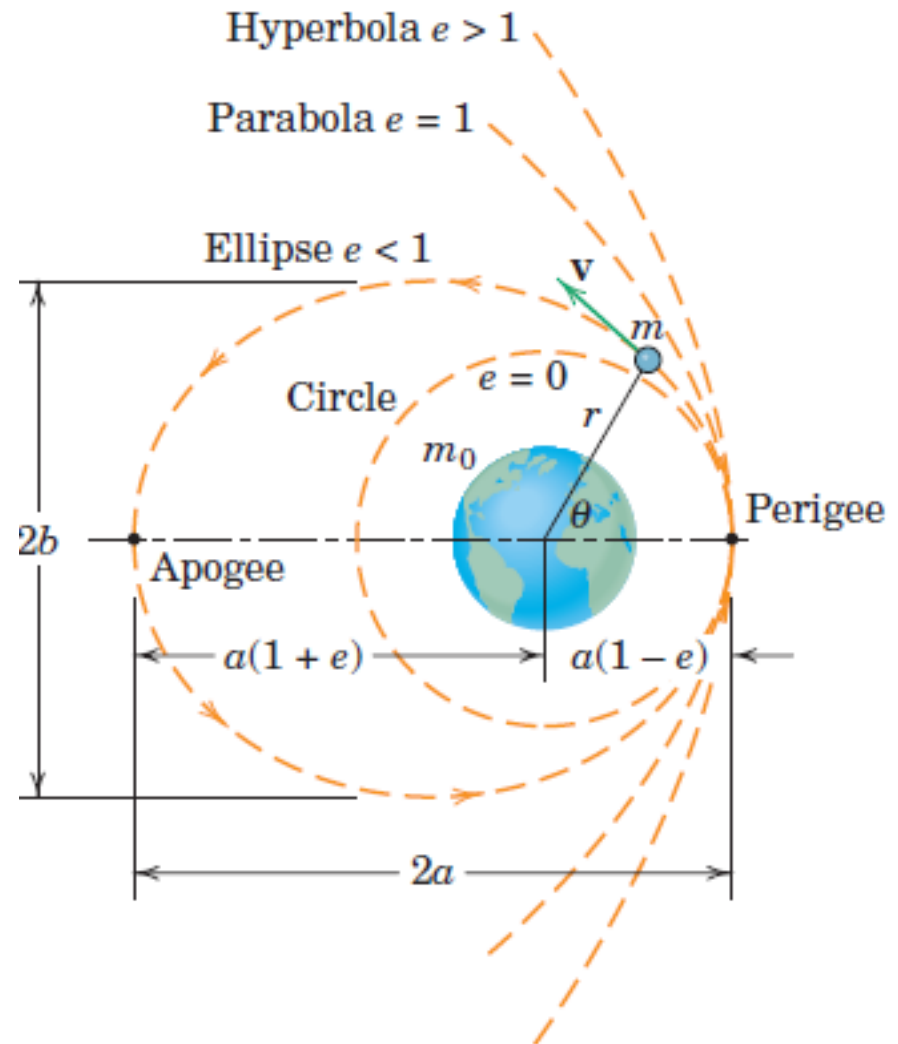
$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d}$$

At $\theta = 0$ $\frac{1}{r_{min}} = \frac{1}{ed} + \frac{1}{d}$

$$r_{min} = \frac{ed}{1 + e}$$

At $\theta = \pi$ $r_{max} = \frac{ed}{1 - e}$



Central Force Motion Ellipse, $e < 1$

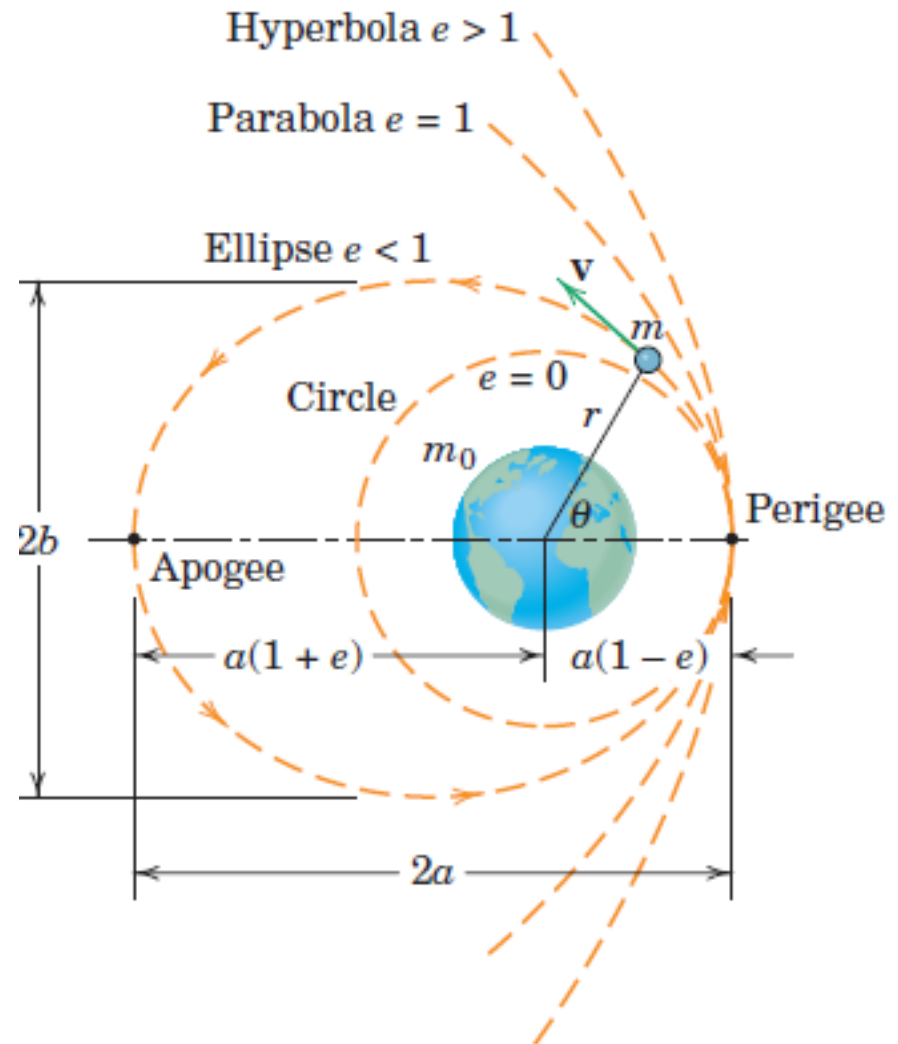
$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d}$$

$$\frac{2a}{1+e} + \frac{2a}{1-e} = \frac{2ed}{1-e^2}$$

$$2a = \frac{2ed}{1-e^2}$$

$$d = \frac{a(1-e^2)}{e}$$



Central Force Motion Ellipse, $e < 1$

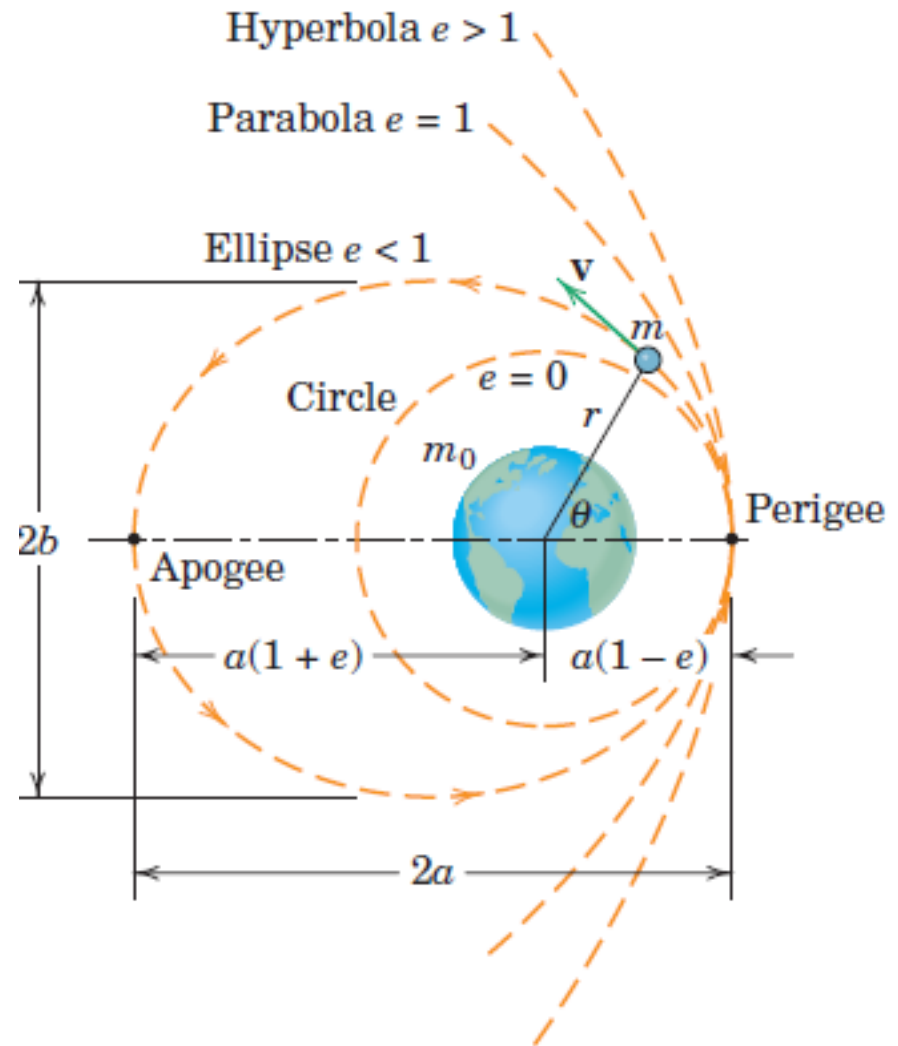
$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d}$$

$$\frac{1}{d} = \frac{e}{a(1 - e^2)}$$

$$\frac{1}{r} = \frac{1}{a(1 - e^2)} + \frac{e \cos\theta}{a(1 - e^2)}$$

$$\frac{1}{r} = \frac{1 + e \cos\theta}{a(1 - e^2)}$$



Central Force Motion Ellipse, $e < 1$

$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

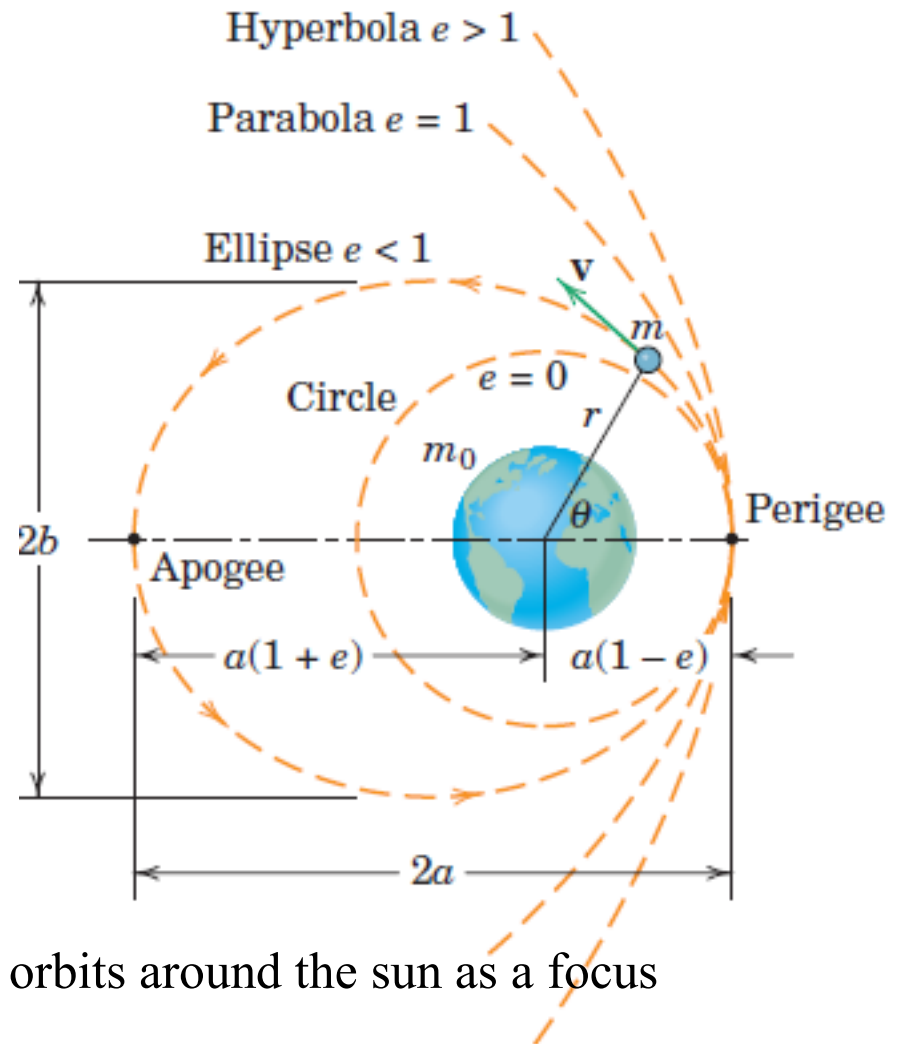
$$\frac{1}{r} = \frac{1 + e \cos \theta}{a(1 - e^2)}$$

At $\theta = 0$

$$r_{min} = a(1 - e)$$

$$r_{max} = a(1 + e)$$

$$b = a\sqrt{1 - e^2}$$

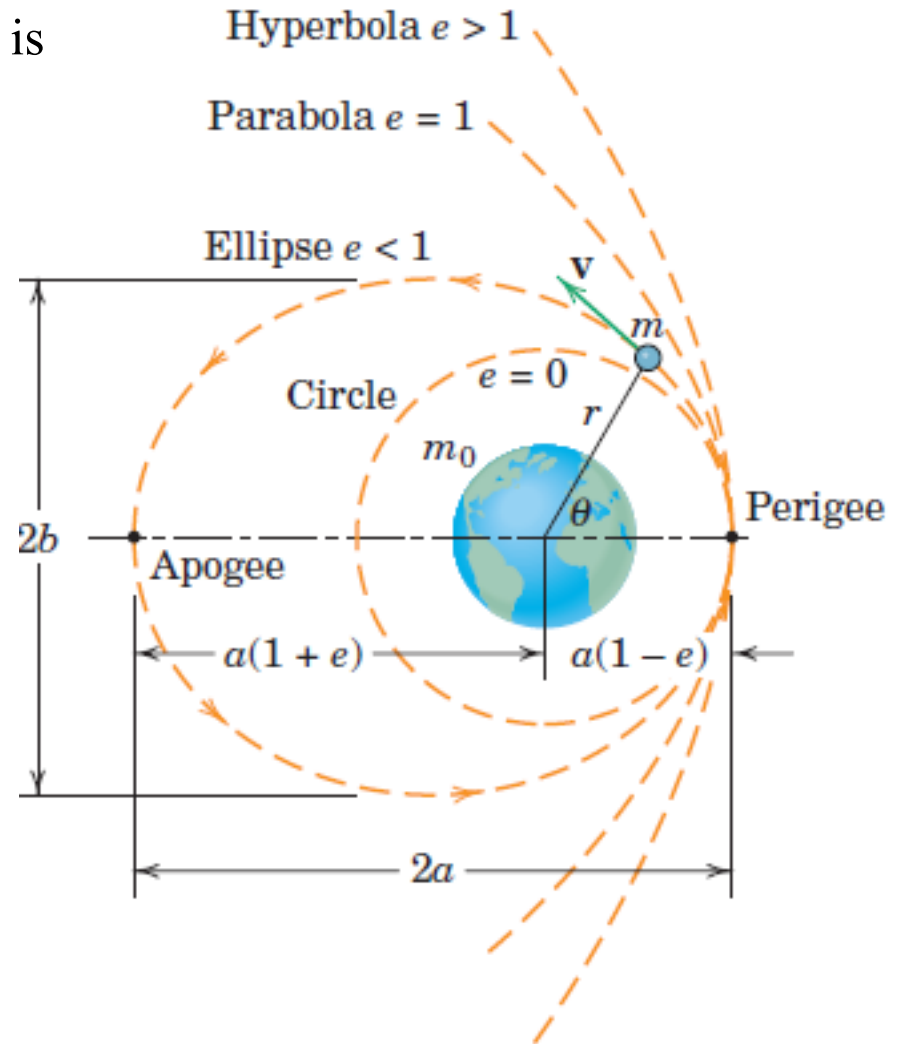


Kepler's first law: Planets move in elliptical orbits around the sun as a focus

Central Force Motion Ellipse, $e < 1$

Time period: Period for the elliptical orbit is the total area of the ellipse divided by the constant rate at which the area is swept through

$$\tau = \frac{A}{\dot{A}} = \frac{\pi ab}{\frac{1}{2} r^2 \dot{\theta}} = \frac{2\pi ab}{h}$$



Central Force Motion Ellipse, $e < 1$

$$\tau = \frac{A}{\dot{A}} = \frac{\pi ab}{\frac{1}{2}r^2\dot{\theta}} = \frac{2\pi ab}{h}$$

$$e = \frac{h^2 C}{Gm_0}$$

$$C = \frac{1}{d}$$

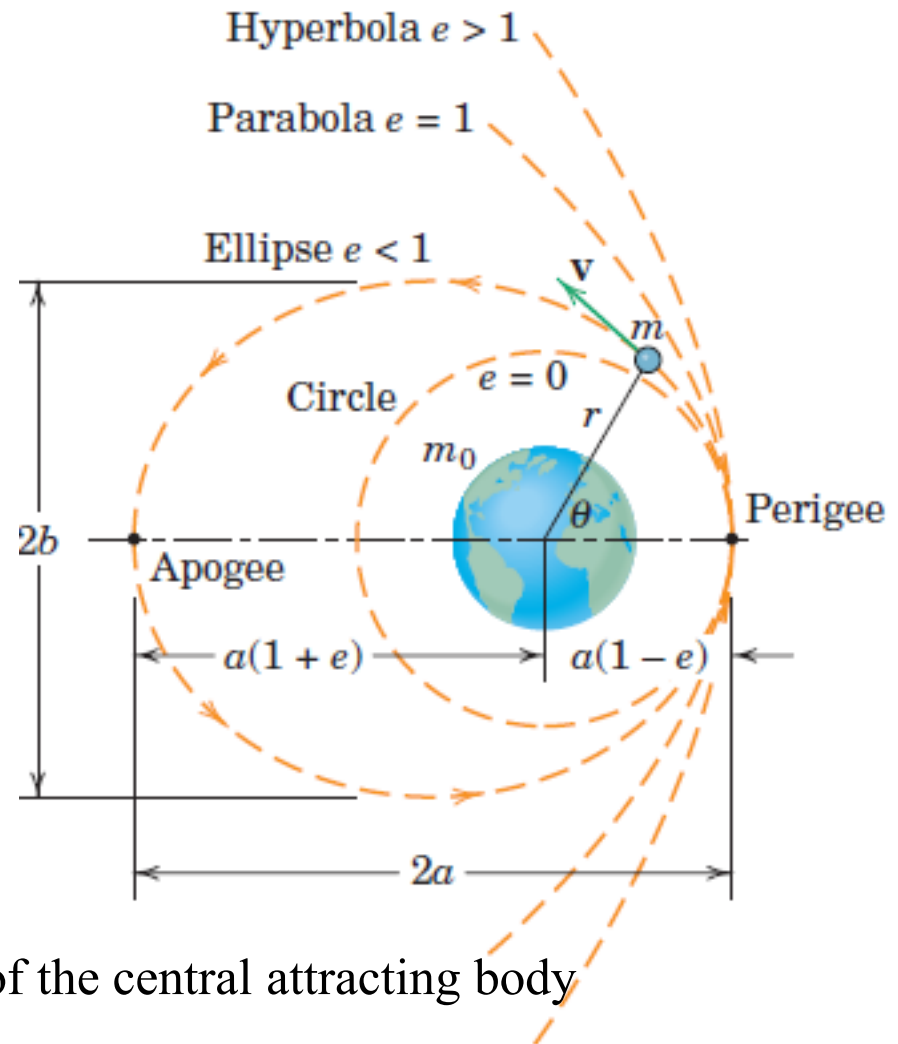
$$Gm_0 = gR^2$$

$$d = \frac{a(1 - e^2)}{e}$$

$$b = a\sqrt{1 - e^2}$$

$$\tau = \frac{2\pi a^{\frac{3}{2}}}{R\sqrt{g}}$$

R is the mean radius of the central attracting body



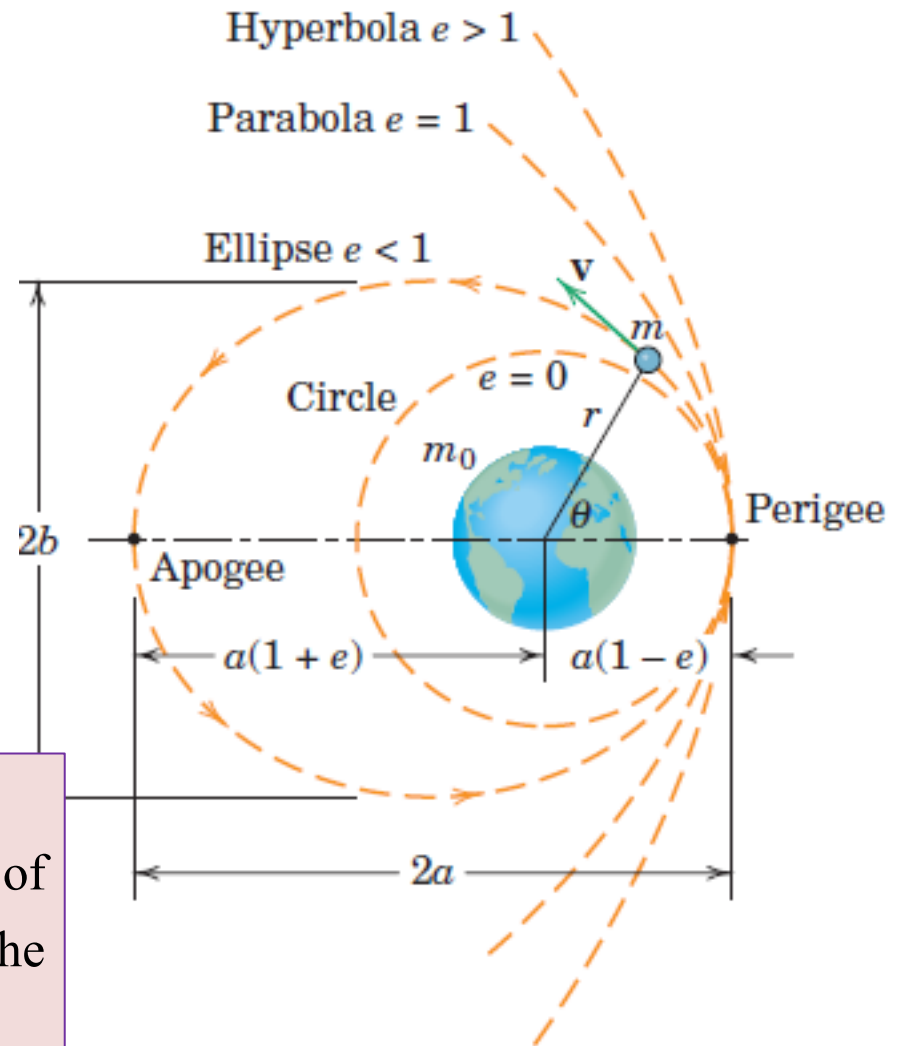
Central Force Motion Ellipse, $e < 1$

$$\tau = \frac{A}{\dot{A}} = \frac{\pi ab}{\frac{1}{2}r^2\dot{\theta}} = \frac{2\pi ab}{h}$$

$$\tau = \frac{2\pi a^{\frac{3}{2}}}{R\sqrt{g}}$$

R is the mean radius of the central attracting body

Kepler's *third law* of planetary motion
which states that the square of the period of motion is proportional to the cube of the semimajor axis of the orbit



Central Force Motion Parabola, $e=1$

$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

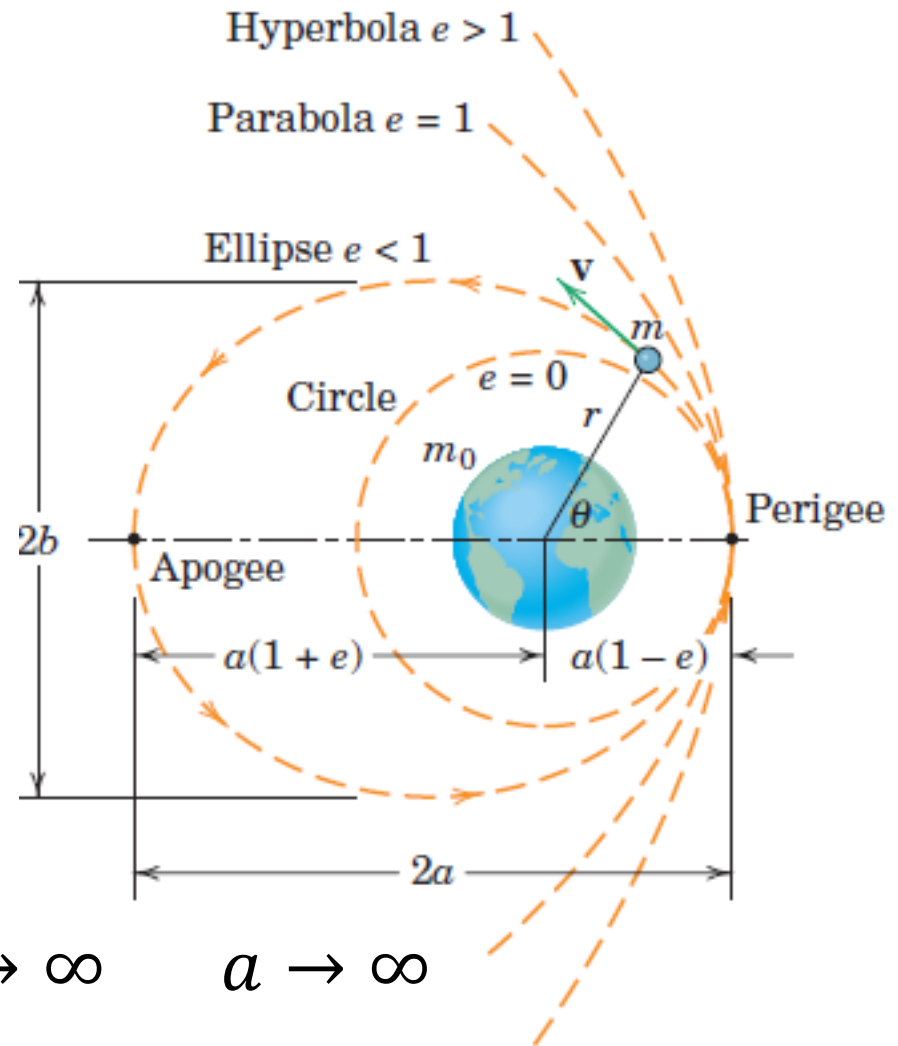
$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d}$$

$$C = \frac{1}{d} \quad ed = \frac{h^2}{Gm_0}$$

$$\frac{1}{r} = \frac{1 + \cos\theta}{d}$$

$$Gm_0 = h^2 C$$

$$\text{At } \theta = \pi \quad \frac{1}{r} = \frac{1 - 1}{d} \quad r \rightarrow \infty \quad a \rightarrow \infty$$



Central Force Motion Hyperbola, $e > 1$

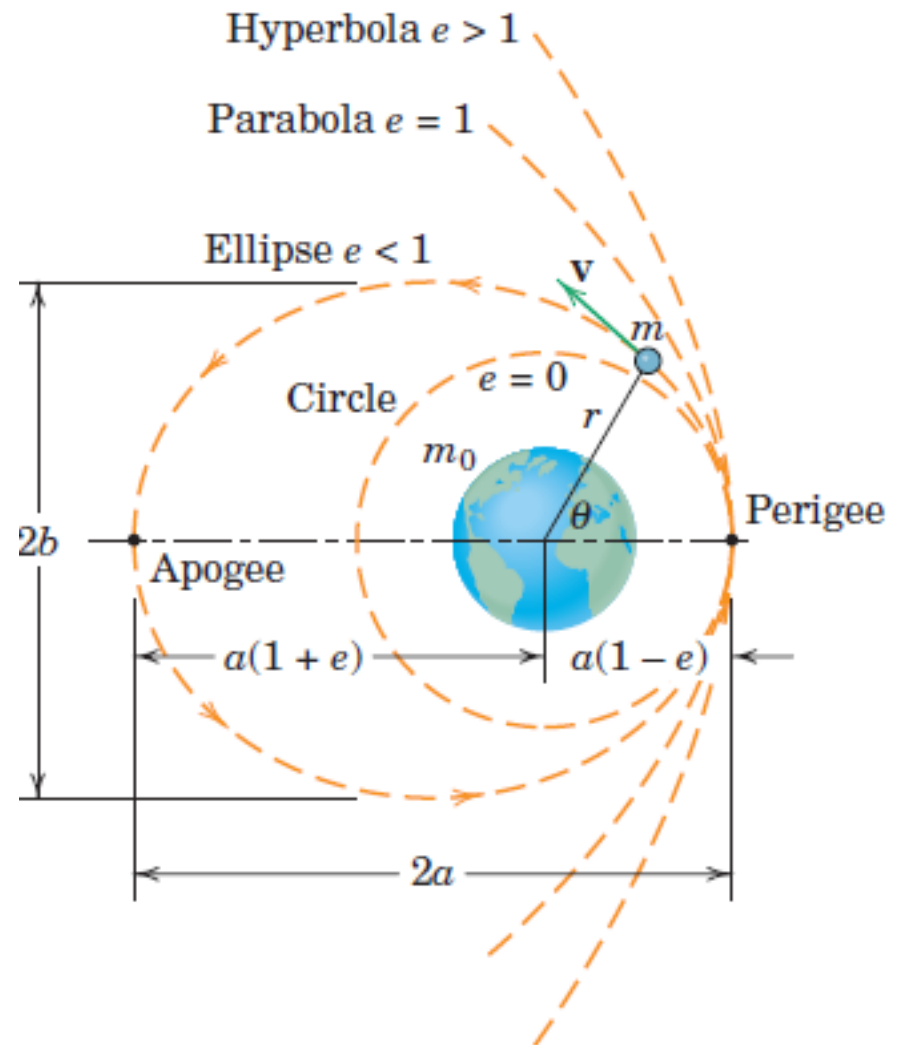
$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d}$$

$$e > 1 \quad \cos \theta_1 = -\frac{1}{e}$$

$$\cos(-\theta_1) = -\frac{1}{e}$$

$$r \rightarrow \infty$$



Central Force Motion Hyperbola, $e > 1$

$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

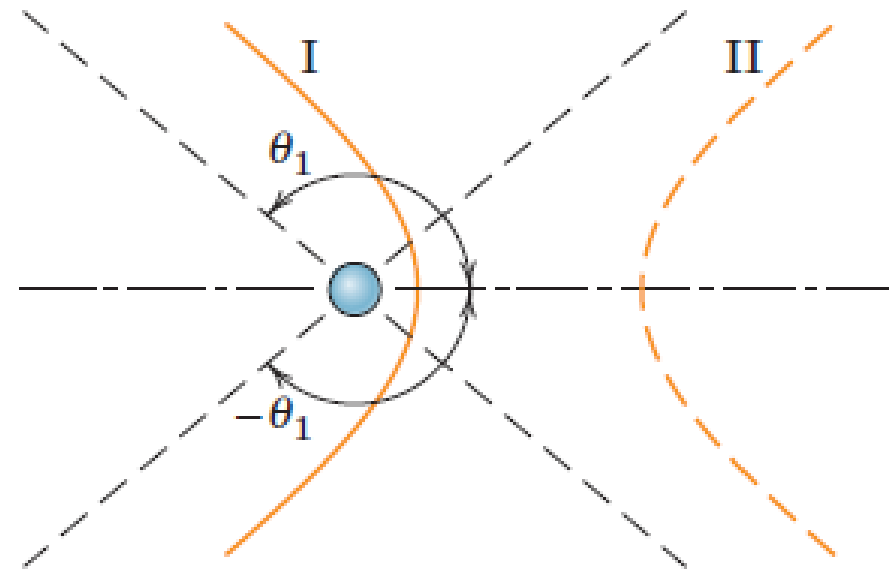
$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d}$$

$e > 1$

$$\cos \theta_1 = -\frac{1}{e}$$

$$\cos(-\theta_1) = -\frac{1}{e}$$

$$r \rightarrow \infty$$



Branch I exists (between angles $-\theta_1$ and θ_1) and branch II (for remaining angles) will lead to negative values of r .

Energy Analysis: Central Force Motion

Conservative System

$$E = T + V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{mgR^2}{r}$$

$$\text{At } \theta = 0 \quad \dot{r} = 0$$

$$\text{At } \theta = 0$$

$$r^2\dot{\theta}^2 = \left(\frac{h}{r}\right)^2$$

$$u = \frac{1}{r} = C \cos(\theta) + \frac{Gm_0}{h^2}$$

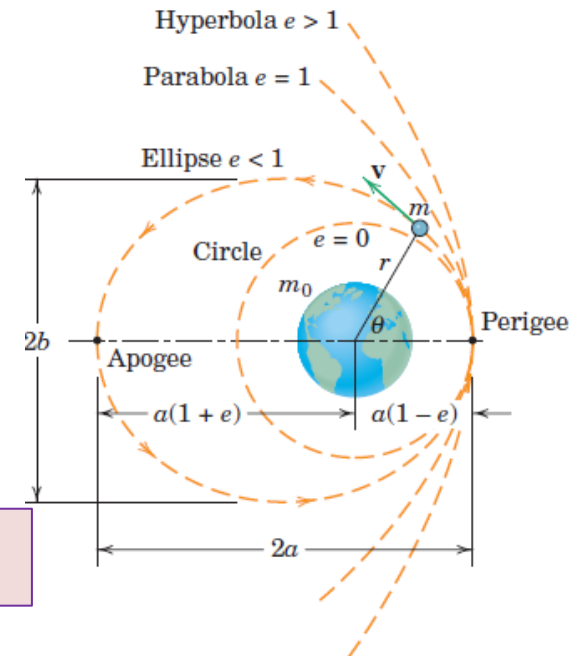
$$\frac{2E}{m} = h^2 C^2 - \frac{g^2 R^4}{h^2}$$

$$e = + \sqrt{1 + \frac{2Eh^2}{mg^2R^4}}$$

$$eGm_0 = h^2 C$$

$$Gm_0 = gR^2$$

$$egR^2 = h^2 C$$



Energy Analysis: Central Force Motion

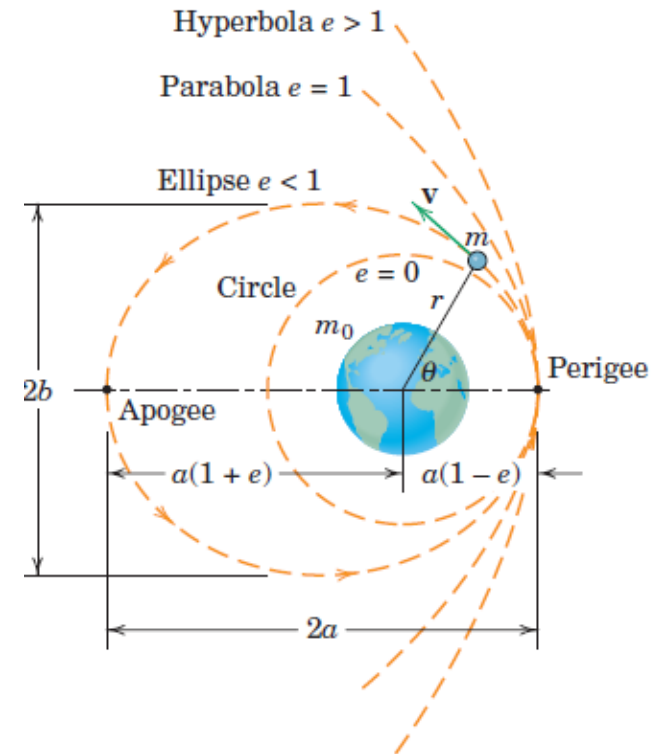
$$e = + \sqrt{1 + \frac{2Eh^2}{mg^2R^4}}$$

$e < 1$ (Ellipse), E is negative

$e = 1$ (Parabola), E is zero

$e > 1$ (Hyperbola), E is positive

These conclusions depend on the arbitrary selection of the datum condition for zero potential energy ($V=0$ when r tends to ∞)



Velocity of Moving Body: Central Force Motion

Conservative System

$$E = T + V = \frac{1}{2}m(v^2) - \frac{mgR^2}{r}$$

$$e = + \sqrt{1 + \frac{2Eh^2}{mg^2R^4}}$$

$$E = -\frac{gR^2m}{2a}$$

$$v^2 = 2gR^2 \left(\frac{1}{r} - \frac{1}{2a} \right)$$

Elliptical Orbit

$$egR^2 = h^2C$$

$$\frac{1}{C} = \frac{a(1 - e^2)}{e}$$

Velocity of Moving Body: Central Force Motion

Conservative System

$$v^2 = 2gR^2 \left(\frac{1}{r} - \frac{1}{2a} \right)$$

$$v_{P(r_{min})} = R \sqrt{\frac{g}{a}} \sqrt{\frac{r_{max}}{r_{min}}}$$

$$v_{A(r_{max})} = R \sqrt{\frac{g}{a}} \sqrt{\frac{r_{min}}{r_{max}}}$$

Elliptical Orbit

$$r_{min} = a(1 - e)$$

$$r_{max} = a(1 + e)$$

$$b = a\sqrt{(1 - e^2)}$$

Assumptions: Central Force Motion

1. The two bodies possess spherical mass symmetry so that they may be treated as if their masses were concentrated at their centers, that is, as if they were particles.
 - ✓ Poor Assumption for: Artificial satellites in the very near vicinity of oblate planets.
 - ✓ Good Assumption for: Bodies which are distant from the central attracting body, Most heavenly bodies
2. There are no forces present except the gravitational force which each mass exerts on the other.
 - ✓ Poor Assumption: Aerodynamic drag on a low altitude earth satellite is a force which usually cannot be ignored in the orbital analysis
3. Mass m_o is fixed in space.
 - ✓ Not so good assumption for earth–moon system. Mass of Moon= 1/81 times mass of Earth

Perturbed Two Body Problem

$$\frac{Gmm_0}{r^3} \mathbf{r} = m_0 \ddot{\mathbf{r}}_1$$

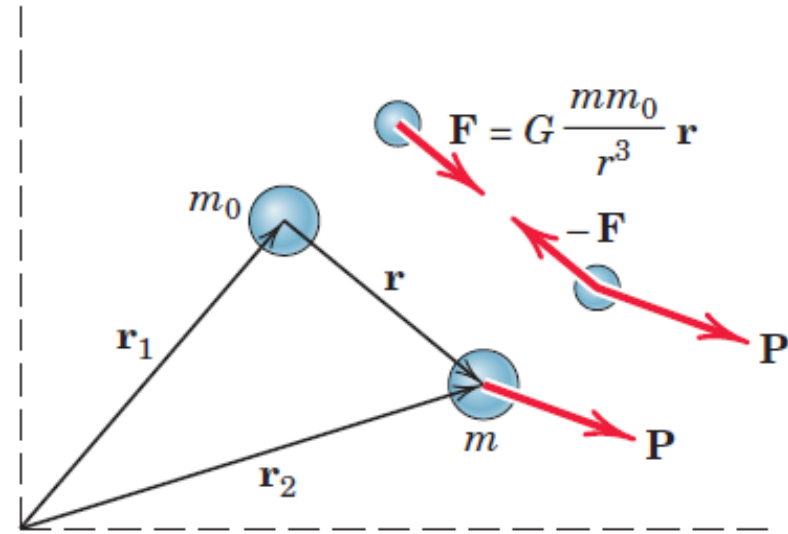
$$-\frac{Gmm_0}{r^3} \mathbf{r} + \mathbf{P} = m \ddot{\mathbf{r}}_2$$

$$-\frac{Gm_0}{r^3} \mathbf{r} + \frac{\mathbf{P}}{m} = \ddot{\mathbf{r}}_2$$

$$\frac{Gm}{r^3} \mathbf{r} = \ddot{\mathbf{r}}_1$$

$$\ddot{\mathbf{r}}_2 - \ddot{\mathbf{r}}_1 = -\frac{Gm_0}{r^3} \mathbf{r} - \frac{Gm}{r^3} \mathbf{r} + \frac{\mathbf{P}}{m}$$

$$\ddot{\mathbf{r}} = -\frac{G(m_0 + m)}{r^3} \mathbf{r} + \frac{\mathbf{P}}{m}$$



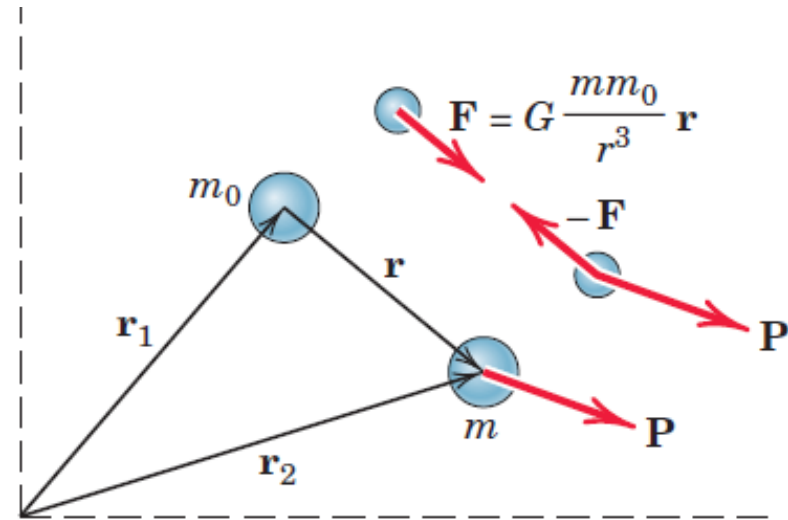
Solution of the equations yields the relative position vector $\mathbf{r}$ as a function of time

Restricted Two Body Problem

$$\ddot{\mathbf{r}} = -\frac{G(m_0 + m)}{r^3} \mathbf{r} + \frac{\mathbf{P}}{m}$$

$$m_0 \gg m \text{ and } \mathbf{P}=0$$

$$\ddot{\mathbf{r}} + \frac{G(m_0)}{r^3} \mathbf{r} = 0$$



$$(\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta + \frac{G(m_0)}{r^2} \mathbf{e}_r = 0$$

$$(\ddot{r} - r\dot{\theta}^2) + \frac{G(m_0)}{r^2} = 0$$

$$(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0$$

Same expressions as derived assuming mass m_0 fixed

Restricted Two Body Problem

$$\ddot{\mathbf{r}} = -\frac{G(m_0 + m)}{r^3} \mathbf{r} + \frac{\mathbf{P}}{m} \quad \text{If } \mathbf{P}=0 \quad \ddot{\mathbf{r}} = -\frac{G(m_0 + m)}{r^3} \mathbf{r}$$

$$m_0 \gg m \text{ and } \mathbf{P}=0 \quad \ddot{\mathbf{r}} + \frac{G(m_0)}{r^3} \mathbf{r} = 0$$

If we replace m_0 by $(m_0 + m)$ in the expressions derived with the assumption of m_0 fixed, then we obtain expressions which account for the motion of m_0

$$\tau = \frac{2\pi a^{\frac{3}{2}}}{\sqrt{Gm_0}}$$

Derived with the assumption of m_0 fixed

$$\tau = \frac{2\pi a^{\frac{3}{2}}}{\sqrt{G(m + m_0)}}$$

With m_0 NOT fixed

THANK YOU